École nationale de la statistique
et de l'analyse de l'information

CAUDULLO, Egle

Non-confidential

MASTER'S THESIS

End-of-Studies Internship

MASTER FOR SMART DATA SCIENCE

HOST ORGANIZATION:

CREST — Center for Research in Economics and Statistics
CREM — Centre de recherche en économie & management

INTERNSHIP SUBJECT:
Measuring cultural polarization across social groups: a new machine learning classification approach

INTERNSHIP LOCATION: ENSAI - 51 Rue Blaise Pascal

CITY: Bruz

COUNTRY: France

Academic Year 2025–2026

Internship Supervisors: François Portier
Emelin Bezin

ENSAI Internship Advisor: François Portier

Contents

1 Introduction **3**
- 1.1 Context . . . 3
- 1.2 The question . . . 3
- 1.3 What we find . . . 4
- 1.4 Related work . . . 4
- 1.5 Structure of the thesis . . . 4

2 Data **5**
- 2.1 Dataset . . . 5
  - 2.1.1 Data Acquisition . . . 5
  - 2.1.2 Data Structure . . . 5
- 2.2 Data Preparation . . . 5
  - 2.2.1 Missing values . . . 5
  - 2.2.2 Data Normalization . . . 5
- 2.3 Descriptive Statistics . . . 6

3 Methodology **7**
- 3.1 Design of the analysis . . . 7
- 3.2 Data representation . . . 8
  - 3.2.1 Principal Component Analysis . . . 8
  - 3.2.2 UMAP . . . 8
  - 3.2.3 Full space . . . 9
- 3.3 Partitioning Algorithms . . . 9
  - 3.3.1 k-means and k-medoids . . . 9
  - 3.3.2 Clustering without a fixed number of groups . . . 10
- 3.4 Deciding how many groups . . . 11
- 3.5 Comparing partitions across methods . . . 11
- 3.6 Characterizing the clusters . . . 12
- 3.7 Recovering membership from observable characteristics . . . 12
- 3.8 Comparing performance across partitions . . . 12

4 Results **14**
- 4.1 There are no discrete value groups . . . 14
  - 4.1.1 Do the three representations capture the same structure? . . . 14
  - 4.1.2 A density-based algorithm leaves most respondents unassigned . . . 17
  - 4.1.3 Disagreement between methods concentrates at the boundaries . . . 18
  - 4.1.4 No selection criterion identifies a number of groups . . . 19
  - 4.1.5 The evidence converges . . . 19
- 4.2 Describing the clusters . . . 20
  - 4.2.1 Choosing k . . . 20
  - 4.2.2 Characterizing the clusters . . . 20
  - 4.2.3 Cluster identities . . . 20
- 4.3 Recovering cluster membership . . . 21
  - 4.3.1 What the exercise measures . . . 21
  - 4.3.2 Politics recovers more than social condition . . . 22
  - 4.3.3 Which cluster is recovered . . . 23

4.3.4 Which observable characteristics carry the signal . . . 24
4.3.5 The density-based partition . . . 25
4.3.6 Observable characteristics tilt a position without fixing it . . . 26

5 Conclusion **28**

A Appendix **31**
A.1 Questionnaire Structure . . . 31
A.2 Missing Value Treatment . . . 32
A.2.1 Filtering and Imputation . . . 32
A.2.2 Limitations . . . 32
A.3 Supplementary Results for the Clustering Stage . . . 33
A.3.1 Interpretation of the Representations . . . 33
A.3.2 Structure Diagnostics . . . 34
A.3.3 Variables Characterising the Clusters at $k = 3$ . . . 35
A.4 Density-Based Clustering across Embedding Dimensions . . . 37
A.4.1 Geometry of the Unassigned Respondents . . . 37
A.4.2 Why so many noise points . . . 37
A.4.3 Parameter Sensitivity . . . 37
A.5 Value Groups as Coalitions . . . 38
A.5.1 The Fractional Hedonic Game . . . 38
A.5.2 Results . . . 39
A.6 Recovery of Cluster Membership at Every k . . . 39

1. Introduction

1.1 Context

Social identity refers to an individual's sense of self derived from perceived membership in social groups, such as race, gender, class, religion, or nationality. Its effect on economic choices has been outlined by Akerlof and Kranton [2000], according to whom a person's social identity enters their objective function directly, and that the choice of a group may be among the most consequential economic decisions anyone makes. People avoid choices that conflict with the identity they hold, even when those choices would pay.

Identity is not observed, so it has to be recovered from something that is. It can be recovered from group membership. *Defining a group is, in this sense, a way of defining an identity.* The empirical problem then becomes concrete: which groups, and on what basis?

Two approaches exist, and this thesis is about the relation between them. First, identity economics usually builds groups from characteristics that can be observed directly and the set of values individuals hold is determined by the group to which each individual belongs. A different approach is proposed by Desmet et al. [2025]: they build groups out of the values themselves, placing together people who answer value questions in similar ways. This last approach leads to a different definition of group membership: not merely social or cultural condition, but it involves people's beliefs. This leads to the hypothesis that one decides which group to belong to on the basis of homophily in values. The number of groups, in their analysis, is fixed at two without a rule that comes from the data. This thesis takes an intermediate position: we construct groups based on individuals' values and then investigate whether, and to what extent, these value-based groups allow us to recover their sociodemographic and political affiliations.

1.2 The question

Two questions are asked here. The first is whether value answers divide Americans into groups at all. Desmet et al. [2025] fix the number of groups at two without a rule that comes from the data, so the prior question — whether the data support any number of groups — is left open by their design, and that is where this thesis starts. For this stage we use only the value questions: political and demographic items are held out, so that no group found at this stage can be an artifact of having asked people their party or their income.

The design is built so that a negative answer cannot be blamed on the tool. The same respondents are described in several different ways and searched with several methods. Each candidate partition is judged not only by the usual internal criteria, which reward compactness within groups and separation between them, but by whether it is a partition its own members would accept: one that nobody would rather leave, in the sense of the fractional hedonic game of Desmet et al. [2025].

The second stage puts the held-out questions back. Taking the first-stage groups as given, we ask how far a respondent's position can be recovered from their politics, from their social condition, and from the two together.

The position defended here is that American value orientations, taken as a whole, are distributed along a continuum rather than divided into groups. Positions on that continuum are legible and can be described, and they are tilted by observable characteristics, politics more than social condition. What they are not is bounded: the population does not separate into camps, and no classifier can recover a boundary the data do not contain.

1.3 What we find

No criterion identifies a number of groups. The distances between respondents form a single smooth distribution with no gap where a boundary would leave one, and the sample has one broad peak rather than several. The method that does not take the number of clusters as input places the groups it does find at the edges of a single connected cloud rather than at its centre. No partition, under any method and any number of groups, is stable: in each of them a substantial share of respondents would rather belong elsewhere.

We nonetheless retain three groups, and state plainly that the choice is interpretive rather than statistical. Three is the largest number for which the same substantive content survives a change of method; beyond it, the additional splits track the choice of algorithm. The three positions should be read as regions of a gradient, not as bounded types.

Recovery from observable characteristics is partial and asymmetric. Political attitudes reconstruct a respondent's position about a third better than social condition does, and combining the two adds little to what politics already provides. Neither approaches the accuracy that discrete, well-separated groups would allow. The same finding is visible one variable at a time, without any model: every age group and every income decile contains a substantial share of all three positions.

1.4 Related work

Three literatures meet here. The first is the one this thesis extends directly: Desmet et al. [2025] recover value groups from survey data and measure the polarisation between them. Their method and its theoretical justification are described above; what we add is the step before it.

The second is Bertrand and Kamenica [2018], who also divide the population in two and use how predictable each group is as a measure of cultural distance. We adopt that logic while relaxing its premise: rather than assuming a division and measuring the distance across it, we take a three-group description whose interpretive status is declared and ask how much of it observable characteristics can reconstruct.

The third is the substantive framework associated with Inglehart and Welzel [2005], in which value change is organised along a small number of axes, the most prominent opposing traditional and secular-rational orientations. Our first principal component reproduces that axis closely. The contribution is not to dispute it but to note what follows from it: an axis is a continuum, and a population distributed along one does not separate into groups merely because the axis has two ends.

1.5 Structure of the thesis

Chapter 2 describes the survey, the treatment of non-response, and the division of the questionnaire into the blocks the analysis keeps apart. Chapter 3 sets out the three representations of the data, the partitioning algorithms, the criteria used to choose a number of groups, and the classifiers used in the second stage. Chapter 4 reports the evidence against discrete groups, describes the three positions retained, and measures how far each block of observable characteristics recovers them. Chapter 5 draws the conclusions and indicates what would be worth doing next. Technical material, robustness checks and the full result tables are in the appendices, cross-referenced from the text at the point where each becomes relevant.

2. Data

2.1 Dataset

2.1.1 Data Acquisition

The data come from the World Values Survey (WVS) Wave 7, United States sample [Haerpfer et al., 2022], archived as `USA_2017_WVS-W7_v01_M`. The WVS is a cross-national research program studying social, political, economic, religious and cultural values, active since 1981 and now comprising seven waves; our analysis focuses on the seventh and involves only American citizens.

Fieldwork for the U.S. sample ran from April to May 2017. The realized sample comprises 2,596 respondents; after removing incomplete records our working sample contains 2,579 observations. From the master questionnaire we retain 198 items covering values, attitudes and normative orientations, excluding demographic and political-culture variables, which are reserved for the supervised characterisation stage.

2.1.2 Data Structure

The raw data follows a standard response-by-question layout: one row per interviewee, one column per questionnaire item ($n = 2{,}596$ original respondents, 290 variables). Each of the 290 questions is answered either on a Likert scale (of three, four, five or ten points) or with a binary answer (yes, no); Appendix A.1 displays the full structure of the questionnaire.

The questionnaire is organized in 14 sections (see Table A.1 in the Appendix), each referring to a subset of the values on which the survey is conducted. We partition the set of questions into three variable blocks, mirroring the two-stage logic of the analysis: unsupervised clustering operates exclusively on the latent-variable block, while political and sociodemographic variables are held out and reintroduced only afterwards, to recover cluster membership. Table A.1 gives, for each section, a brief description of the questions and their number.

2.2 Data Preparation

2.2.1 Missing values

The WVS encodes non-response with a set of reserved codes (-1 = don't know, -2 = no answer/refused, -3 = not applicable, -5 = missing for other reasons), which are recoded as `NA` before any analysis. Following Desmet et al. [2025], we then apply two sequential response-rate filters — dropping variables answered by fewer than 30% of respondents, and respondents answering fewer than 70% of the remaining variables — and impute the residual missingness with a Random Forest classifier trained on the subset of fully complete respondents. This leaves a working sample of $n = 2{,}579$ observations and 285 variables with no remaining missing entries. The removed questions belong to the sociodemographic block, so their removal does not affect the clustering stage of our analysis. The full procedure, together with our departures from Desmet et al.'s treatment and its main limitations, is documented in Appendix A.2, where Table A.2 breaks non-response down by section and by type.

2.2.2 Data Normalization

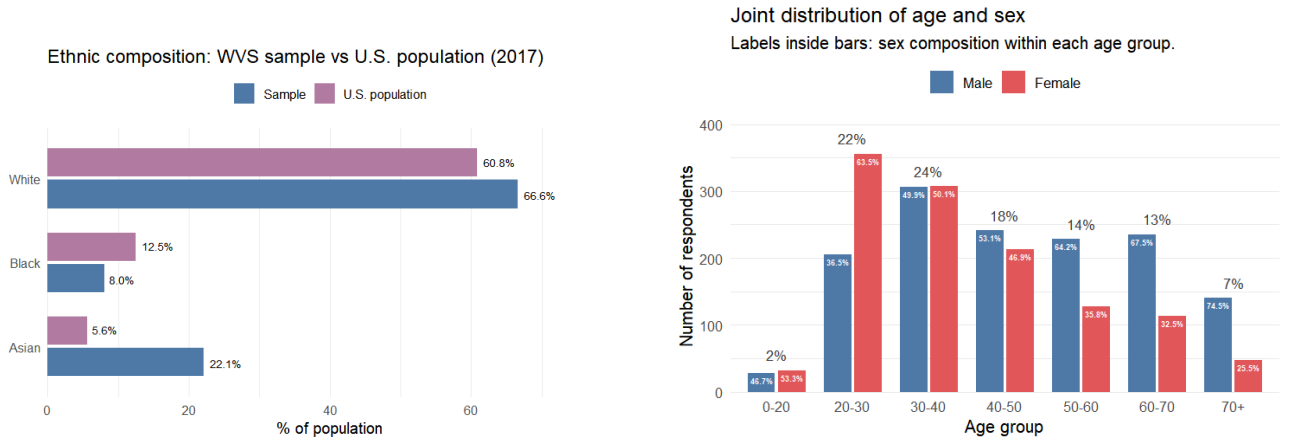
The latent value variables Z_i are measured on heterogeneous scales. Since the starting point of our pipeline operates on Euclidean distances, and distance is not invariant to the units in which each coordinate is expressed, this heterogeneity must be resolved before any geometry is imposed on the data.

Explicit standardisation (PCA, UMAP) Both the principal component decomposition and the UMAP embedding are preceded by the same z-score standardisation, so that every item enters with unit variance. This is where our treatment departs from Desmet et al. [2025]. They do not standardise to unit variance; instead they rescale every variable linearly to the unit interval $[0, 1]$. This equalises *ranges* but not *variances*, and the two are far from equivalent for survey data.

Normalisation internal to the metric (Gower) For the k-medoids implementation (through the PAM algorithm, discussed in Section 3.2.3), the question of scaling does not arise separately, because it is absorbed into the choice of Gower dissimilarity as a metric, which is bounded in $[0, 1]$ by construction.

2.3 Descriptive Statistics

Our sample is composed of $2{,}579$ observations which should represent the American population; Figure 2.1 describes its main characteristics.



(a) Ethnic composition, sample versus U.S. population.

(b) Joint distribution of age and sex.

Figure 2.1: Demographic composition of the WVS sample.

Panel (a): population shares are 2017 resident-population estimates [National Center for Education Statistics, 2021, Table 101.20] and refer to all ages, whereas the survey covers adults aged 18 and over. Categories absent from either source are omitted. The instrument aggregates South and East Asian respondents into a single *Asian* category to match the reference classification. Note that Q290 offers no Hispanic category, although persons of Hispanic origin account for 18.0% of the U.S. population in 2017 and are counted separately from race in the reference source; respondents of Hispanic origin were therefore obliged to select one of the listed categories. Panel (b): percentages above each pair of bars give the share of the sample falling in that age group, while labels inside the bars give the sex composition within the group. Respondents aged 30–40 and 20–30 are the two largest groups (24% and 22%), the 0–20 group being residual (2%) since the survey covers adults only. The sex balance reverses over the life cycle: women outnumber men below the age of 40, whereas men make up roughly two thirds of respondents aged 50 and over. This reversal runs counter to the female-majority sex ratio observed at older ages in the U.S. population and is therefore likely to reflect the sampling design rather than the underlying demography.

3. Methodology

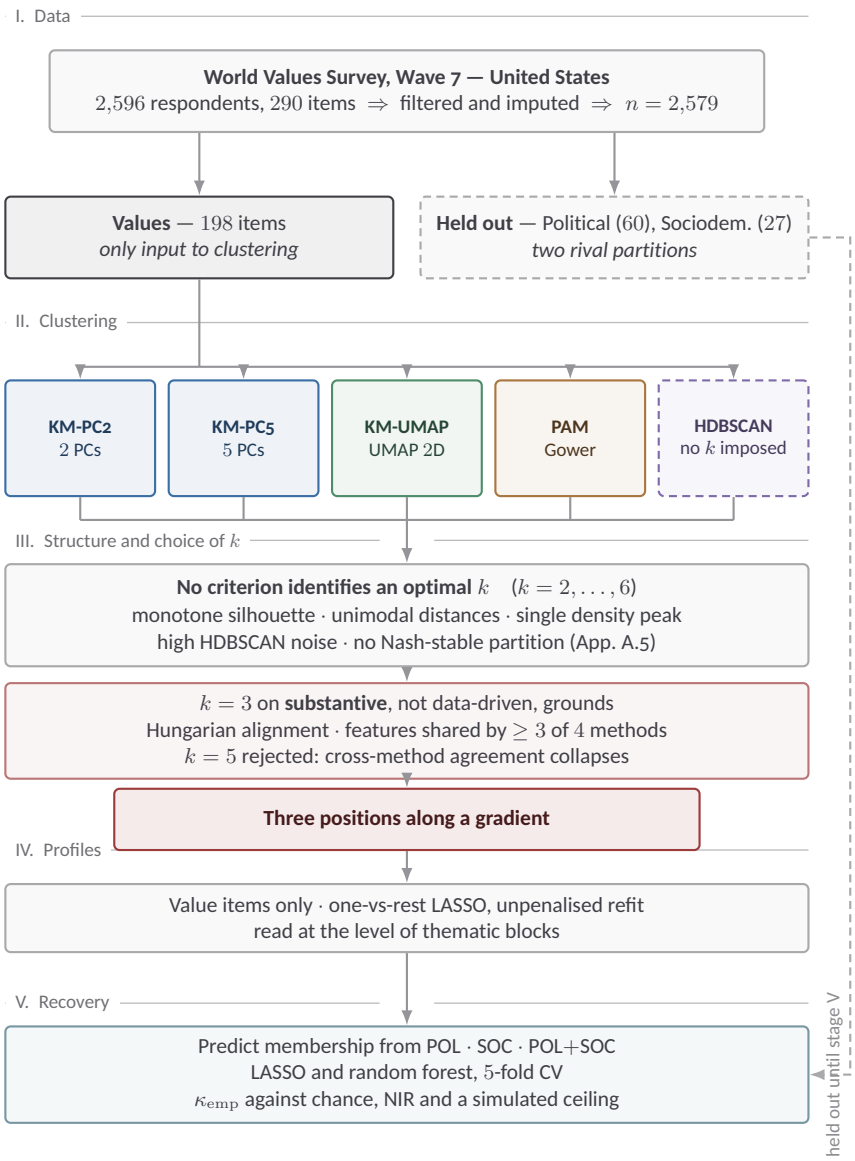


Figure 3.1: Analytical pipeline. Only the value block enters the unsupervised stage; the political and sociodemographic blocks are held out and reappear in the final step as two rival partitions of the same respondents. Four partitioning configurations are estimated, with HDBSCAN alongside as a density-based check that imposes no k. Across $k = 2, \ldots, 6$ no internal criterion isolates an optimum, which is itself the substantive finding. The value $k = 3$ is therefore retained on interpretive grounds, and the final stage measures how far membership in a value group can be reconstructed from each of the other two partitions.

3.1 Design of the analysis

Figure 3.1 summarises the pipeline. Let $\mathbf{Z} \in \mathbb{R}^{n \times p}$ denote the data matrix where n is the number of observations and p the number of variables. We face three problems in turn: (i) finding the appropriate space in which to represent the data, (ii) applying the appropriate clustering algorithm to find the latent groups derived from the

value-based questions, and (iii) recovering cluster membership from observable characteristics. To address the first problem we follow three different paths: principal component analysis, UMAP, and a native-space representation through the Gower distance. For the partitioning phase we apply the k-means algorithm, the Partitioning Around Medoids (PAM) algorithm, and HDBSCAN. To recover cluster membership we use both linear methods (LASSO, logit) and a non-linear one, the random forest.

The analysis is built so that a negative result cannot be blamed on the tool. If no group structure is found, the natural objection is that a different method would have found one. We therefore describe respondents in three ways, apply to each the partitioning algorithm its geometry admits, and add a fourth algorithm that is free to report that no groups exist. A conclusion surviving all of these cannot be attributed to any one of them.

The methods are ordered by increasing complexity, each addressing a limitation of the previous one. Two principal components are the simplest representation and the one used by Desmet et al. [2025]; five components retain more variation (and present some differences with respect to the two-component framework); UMAP relaxes linearity; HDBSCAN removes the requirement to fix the number of groups in advance. We also present results for the Gower dissimilarity, which abandons dimensionality reduction and works on the data in their native mixed-type form; this choice is made so that representation-specific biases are avoided. The substantive conclusion does not change along this sequence, which is why the sequence is constructed.

The value items and the covariates are kept strictly apart. Clustering uses only the $p = 198$ value items Z_i. The 60 political and 27 sociodemographic items X_i take no part in it and re-enter only in Section 3.7, as predictors of a partition they did not help produce.

3.2 Data representation

3.2.1 Principal Component Analysis

Principal component analysis provides the simplest numerical representation and the one permitting comparison with the reference study. Each item is standardised to zero mean and unit variance, and respondents are projected onto the leading eigenvectors of the sample covariance matrix.

PCA seeks a sequence of orthogonal directions of maximum variance in $\mathbf{Z}$. The m-th principal component is defined by the unit loading vector $\boldsymbol{\phi}_m \in \mathbb{R}^p$, orthogonal to all previous components, that maximises the variance of the projected scores. The score of observation i on component m is

$$s_{im} \;=\; \boldsymbol{\phi}_m^{\top}\mathbf{z}_i, \qquad i = 1, \ldots, n, \quad m = 1, \ldots, M,$$

where the loading vector $\boldsymbol{\phi}_m$ is the eigenvector of the sample covariance matrix $\widehat{\boldsymbol{\Sigma}}$ associated with its m-th largest eigenvalue λ_m. The individual loadings ϕ_{jm} quantify the contribution of variable j to the m-th axis of variation.

The scaling choice departs from the reference study. Desmet et al. [2025] rescale each variable linearly to the unit interval, which equalises ranges but not variances; we standardise to unit variance. The two are not equivalent for survey data, since a ten-point scale and a binary item have very different variances even after their ranges are matched.

Two dimensionalities are retained. Two components allow direct comparison with the reference study and admit planar visualisation; five components retain more variation and check that the results are not an artefact of severe truncation. Both are carried through the analysis as separate configurations, KM-PC2 and KM-PC5.

3.2.2 UMAP

UMAP is included because it is biased towards finding groups. The algorithm builds a nearest-neighbour graph in the original space and seeks a low-dimensional layout preserving it, minimising the cross-entropy between the two representations. The low-dimensional edge weights follow a heavy-tailed kernel, which allows a low-dimensional representation without the distances collapsing.

The objective therefore offers little resistance to opening voids between point clouds, and the minimum-cost configuration is fragmented even when the underlying data are continuous.

This makes UMAP a powerful instrument for the question. A failure to find groups under a method that manufactures them is stronger evidence than a failure under a neutral method. We configure it with `n_neighbors` $= 15$, `min_dist` $= 0.1$ and Euclidean metric on the standardised items.

3.2.3 Full space

Alongside the PCA- and UMAP-based approaches, we include a clustering configuration that operates directly on the full $p = 198$-dimensional variable space using a distance measure specifically suited to mixed-type data. Unlike Euclidean distance, Gower accommodates continuous, ordinal, and binary variables within a single framework without requiring standardisation assumptions. Since K-means centroids are not defined under a dissimilarity-only representation, this choice requires applying K-medoids (PAM), which, instead of taking cluster means as centroids, selects actual observations as representatives and minimises the average dissimilarity within each cluster. This configuration differs from the PCA-based approaches along two dimensions simultaneously: (i) distance metric and (ii) absence of dimensionality reduction. Comparing this representation with the other methods allows us to partially disentangle the effect of each choice on the resulting partition.

$$d_{\mathrm{G}}(z_i, z_{i'}) = \frac{\sum_{k=1}^{p} w_k \, \delta_{ii'k} \, d_{ii'}^{(k)}}{\sum_{k=1}^{p} w_k \, \delta_{ii'k}}, \qquad d_{ii'}^{(k)} = \begin{cases} \dfrac{|z_{ik} - z_{i'k}|}{R_k} & \text{quantitative or ordinal,} \\ \mathbb{1}\{z_{ik} \neq z_{i'k}\} & \text{nominal,} \end{cases} \tag{3.1}$$

where R_k is the observed range of item k, w_k its weight (one throughout), and $\delta_{ii'k}$ an indicator that the comparison is admissible for both respondents. Each partial dissimilarity lies in $[0, 1]$ by construction, so normalisation is internal to the metric and no external standardisation is required.

3.3 Partitioning Algorithms

After projecting the data onto a two-dimensional space in which Euclidean distances are meaningful, we proceed to partition the observations into clusters. This stage raises two main methodological questions: (i) which algorithm should be used to construct the clusters, and (ii) how should the optimal number of clusters be determined?

3.3.1 k-means and k-medoids

Both partitioning algorithms minimise the same objective and differ only in where the representative is allowed to live. For a partition $\mathbf{C} = \{C_1, \ldots, C_K\}$,

$$J(C_1, \ldots, C_K) = \sum_{i=1}^{K} \sum_{z \in C_i} D(z, r_i), \qquad r_i = \arg\min_{\tilde{z} \in \mathcal{R}_i} \sum_{z \in C_i} D(z, \tilde{z}), \tag{3.2}$$

where D is a cost function and $\mathcal{R}_i$ the domain of the representative. Table 3.1 reports the two instances.

Table 3.1: The two instances of the common objective (3.2). The methods differ not only in the cost function but in the optimisation domain, and the second difference is the one imposed by the data.

	k-means	k-medoids (PAM)
Cost function $D(z, \tilde{z})$	$\|z - \tilde{z}\|_2^2$	$d_{\mathrm{G}}(z, \tilde{z})$
Domain $\mathcal{R}_i$	$\mathbb{R}^p$ (continuous)	C_i (discrete)
Representative r_i	$\boldsymbol{\mu}_i = \dfrac{1}{\lvert C_i \rvert} \displaystyle\sum_{z \in C_i} z$	$m_i \in C_i$, no closed form
Admissible data	numerical	mixed-type

For k-means the inner minimisation admits a closed-form solution. Differentiating the within-cluster sum of squared Euclidean distances over $\mathbb{R}^p$ yields the arithmetic mean of the cluster members, a point that need not belong to the dataset. Lloyd's algorithm is run from $1{,}000$ random starts and the solution with lowest within-cluster variance is retained.

For PAM the minimisation is discrete and has no closed form. The medoid is constrained to be an actual respondent, $m_i = \arg\min_{\tilde{z} \in C_i} \sum_{z \in C_i} d_{\mathrm{G}}(z, \tilde{z})$, and the minimum is found by enumeration over C_i.

k-means is retained for a reason beyond convention. Its objective admits the game-theoretic reading required in Section 3.4, in which membership is treated as the outcome of individual choices rather than as a partition imposed from outside. This is what allows the stability criterion to be posed at all.

Why d_{G} is not an ℓ_1 norm. Were every variable quantitative or ordinal, (3.1) would reduce to an ℓ_1 distance on range-normalised variables, divided by p, and the two methods would indeed differ only in the exponent of the norm. This is not the case here. A subset of the retained items — among them the type of social change (Q42), the postmaterialist choice batteries (Q152–Q157) and the two questions on the meaning of religion (Q174–Q175) — offer alternatives with no natural ordering, and for these the partial dissimilarity is the indicator in (3.1), which carries no metric content: the dissimilarity between *revolutionary change* and *gradual reform* equals that between *revolutionary change* and *defending the present society*. No norm on $\mathbb{R}^p$ reproduces this behaviour, and the arithmetic mean in (3.2) is undefined for such variables. The discreteness of $\mathcal{R}_i$ is imposed by the measurement level of the data.

3.3.2 Clustering without a fixed number of groups

Every method described so far must be told how many groups to find, which prejudges the question. HDBSCAN does not. It builds a hierarchy of density levels, extracts the clusters persisting longest across that hierarchy, and leaves unassigned any point in a region of insufficient density. The number of groups is an output, not an input.

Construction

The algorithm rests on a single reweighting of the distance. The core distance of a point is the distance to its `minPts`-th nearest neighbour,

$$d_{\mathrm{core}}(x) = d\big(x, \texttt{minPts}\text{-th nearest neighbour}\big), \qquad d_{\mathrm{mreach}}(x, y) = \max\{d_{\mathrm{core}}(x),\, d_{\mathrm{core}}(y),\, d(x, y)\}, \tag{3.3}$$

and a minimum spanning tree is built on d_{mreach} by single linkage. Clusters are then extracted from the condensed hierarchy by persistence.

The reweighting exists to defeat chaining. Single linkage merges, at each step, the two components whose closest pair is nearest, so a thin trail of points connecting two dense regions causes the rule to walk along it and fuse them: one filament is enough to collapse two modes. The mutual reachability distance corrects this by inflating exactly those filaments, since the core distance is large not where points are few but where they are not *concentrated*.

The algorithm's output includes the possibility of unassigned points, or noise points. These points are left out when the minimum spanning tree is split and, after an edge is cut, fewer than `minPts` points remain on its side.

Choice of `minPts`

The single granularity control requires justification because it governs three things at once. In the R implementation `minPts` sets the core-distance neighbourhood, and hence the smoothness of the density estimate; it enters the graph construction through (3.3); and it fixes the minimum cluster size, so it also determines when a branch pruned from the tree becomes noise rather than a cluster.

We take $\texttt{minPts} = 25$ as the primary setting and report a full sweep. With $n = 2{,}579$, this requires a cluster to contain roughly one per cent of the sample to be retained, which is a mild requirement, while making the density estimate smooth enough that isolated respondents do not each become their own mode. We sweep $\texttt{minPts} \in \{5, 10, 15, 20, 25, 30, 40, 50\}$ and summarise the sweep in Appendix A.4.3.

The algorithm is run at four embedding dimensionalities. HDBSCAN is known to degrade in high dimension, but two dimensions may be too lossy for clustering as opposed to visualization, so we run $d \in \{2, 5, 20, 50\}$ and assess agreement between the resulting partitions by the Adjusted Rand Index (Table A.9).

3.4 Deciding how many groups

The number of groups is chosen by a procedure fixed in advance. Three criteria are applied. If none identifies an optimum, the number is chosen on substantive grounds by the rule stated at the end of this section, and the partition is reported as a description rather than as a discovery.

The first criterion is **internal validity.** Average silhouette width measures how much closer each respondent is to its own cluster than to the nearest alternative, and a genuine grouping produces a local maximum at the correct number. We compute it for $k = 2, \ldots, 6$ in each representation; its definition is given in Appendix A.3.2.

Nash equilibrium. The second criterion asks whether a partition survives individual defection. Following Karakaya [2011], the partition is read as a fractional hedonic game: each respondent derives utility from the average similarity to the other members of their group, and the partition is Nash-stable if no individual would raise their utility by moving. This criterion is demanding in a way the others are not, because it evaluates the groups from the point of view of their members rather than from that of an external objective function. It is also the criterion that connects the statistical exercise to the substantive claim: a value community whose members would rather belong elsewhere is not a community, assuming that their choice depends only on the group's members. A formal description is given in Appendix A.5.

Robustness of the results as a criterion. Should no criterion identify an optimum, we retain the largest k for which the same thematic core remains salient across the different clustering configurations (KM-PC2, KM-PC5, KM-UMAP and PAM-Gower). The rule is interpretive and we do not present it otherwise. Each cluster of each clustering configuration is used as the response variable of a one-vs-rest LASSO logistic regression. The selected k is the one for which we observe the highest agreement on the retained blocks of variables across the different methodologies.

3.5 Comparing partitions across methods

Cluster labels are arbitrary, so partitions must be aligned before comparison. We align each partition to a common reference by the Hungarian algorithm, which finds the label permutation maximising agreement (Appendix A.3.2), and report the residual discordance as the share of respondents assigned differently after alignment.

3.6 Characterizing the clusters

Clusters are described only by features surviving a change of method. For each of the four configurations, a one-versus-rest LASSO logistic regression is fitted per cluster on all value items, and the twenty strongest regressors are refitted without penalty to obtain odds ratios. The core of each profile is made of the variables selected by at least three of the four configurations; variables selected by two are kept only as supporting detail, and those selected by a single configuration are discarded. The reported effect is the geometric mean of the odds ratios across the configurations selecting the variable. The variables retained are listed in Table A.7 (Appendix A.3.3).

The encoding matches the treatment used in clustering. Ordered scales enter as single numeric scores, consistent with the ordinal branch of (3.1); unordered items are dummy-coded against their neutral category, so an odds ratio reads as a deviation from taking no position.

Two properties of the LASSO require the results to be read thematically rather than item by item. The penalty selects arbitrarily among collinear predictors, and the religiosity block alone contains some seventeen closely related items, so which one is retained is not stable across resamples. The p-values from the unpenalised refit are descriptive only, since the regressors were selected on the same data.

3.7 Recovering membership from observable characteristics

The second question of the analysis is whether a respondent's position can be recovered from observable features (such as sociodemographic variables). Here cluster membership becomes the response, predicted from covariates that took no part in the clustering.

Three predictor blocks are used, and the comparison between them is the point. The political block alone tests whether value position is a restatement of partisanship; the sociodemographic block alone tests whether it is a restatement of social condition; the two combined show whether they carry overlapping or complementary information. Continuous covariates are binned before entering the design matrix, for interpretative reasons.

Three classifiers span the range from linear to non-parametric. Multinomial logistic regression is the linear baseline; its LASSO-penalised version adds variable selection; a random forest of 300 trees provides a non-parametric benchmark. All are evaluated by 5-fold cross-validation with folds stratified by cluster, so every reported figure is measured out of sample.

3.8 Comparing performance across partitions

Raw accuracy cannot be compared across partitions with different numbers of clusters. The accuracy attainable by chance falls mechanically as clusters multiply — 50% at $k = 2$, 33% at $k = 3$, 25% at $k = 4$ — so the same figure carries different information at different k. We therefore report Cohen's κ [Cohen, 1960],

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{3.4}$$

where p_o is observed accuracy and p_e the accuracy expected by chance. The measure equals one for perfect recovery, zero at chance level, and is negative for systematic disagreement.

The interpretation departs from the original one. Cohen's κ measures agreement between two raters classifying the same items independently, and the independence assumption underlying p_e holds by construction. Here the two "raters" are the fitted model and the cluster labels, and the model was trained on those labels. The question is accordingly not how far two observers agree but whether the covariates contain enough information to reconstruct membership, and we use κ as an index of separability in that sense.

Aggregate figures are supplemented by per-cluster recall. A global accuracy of sixty per cent is consistent with one cluster being recovered almost perfectly and another not at all, and the distinction matters when clusters

differ greatly in size. We therefore report, for each partition, the share of each cluster’s true members correctly identified.

4. Results

4.1 There are no discrete value groups

Every diagnostic points the same way: American value orientations are distributed continuously. This section reports evidence from distances, from density, from a density-based algorithm allowed to find nothing, and from the standard selection criteria. These rest on different assumptions and could have disagreed.

4.1.1 Do the three representations capture the same structure?

Distances between respondents form a single smooth distribution

If respondents fell into groups, the distances between them would form two regimes. Pairs within a group would be close and pairs across groups far, producing a distribution with two modes and a trough between them. Figure 4.1 shows the observed distributions under all three representations, and none has a secondary mode.

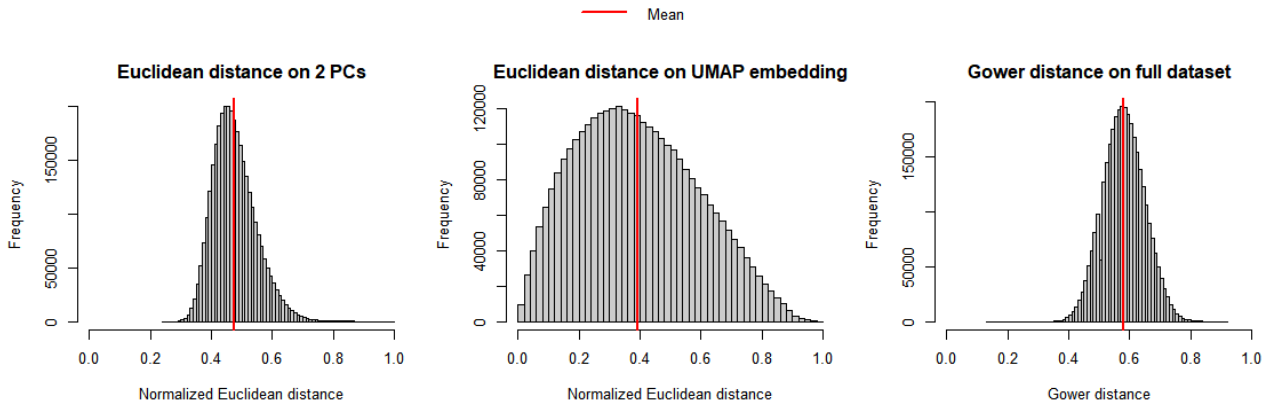


Figure 4.1: Distributions of pairwise distances between respondents under the three representations: Euclidean distance on the first two principal components (left), Euclidean distance on the two-dimensional UMAP embedding (centre), and Gower distance on the full set of 198 value items (right). Euclidean distances are rescaled to $[0, 1]$ by division by the maximum observed distance so that the panels are comparable; Gower distance is bounded in $[0, 1]$ by construction. The red line marks the mean of each distribution. All three are unimodal and concentrated around their mean. A partition into groups would produce a second mode, corresponding to between-group pairs; none is present.

The UMAP distribution is the most informative of the three. Its tails are heavier, which is the expected consequence of the geometry described in Section 3.2.2, and it is therefore the representation most likely to display separation. No separation into two regimes appears there either.

The density surface has a single dominant peak

The same conclusion holds when respondents are viewed as a density rather than as a set of distances. Figure 4.2 shows kernel density surfaces over the three two-dimensional embeddings. The PCA and MDS surfaces show one broad peak; the UMAP surface shows the secondary ridges its construction favours, and Section 4.1.2 shows what happens when a density-based algorithm is asked to treat those ridges as clusters.

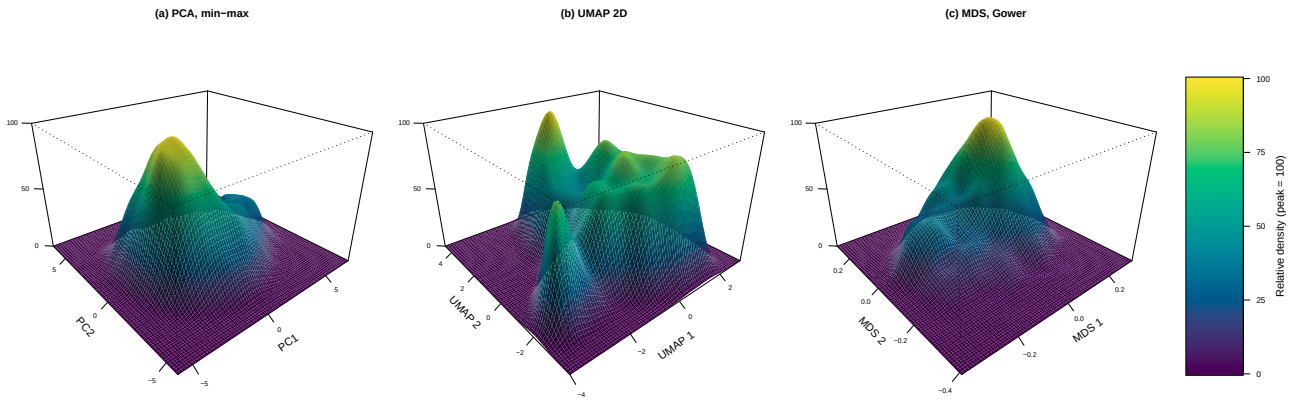
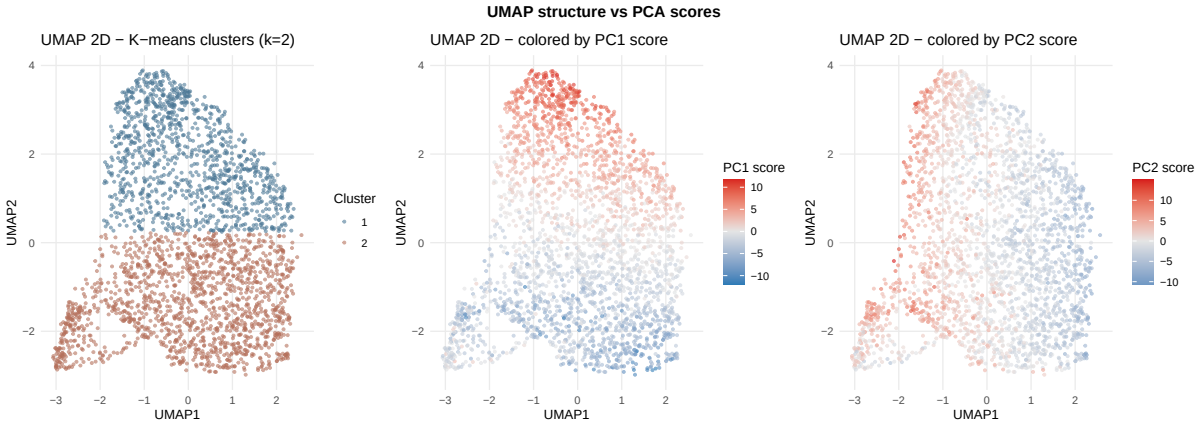
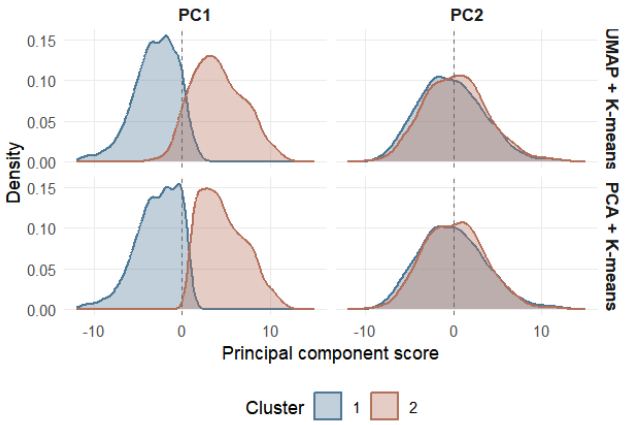


Figure 4.2: Kernel density surfaces over three two-dimensional embeddings of the same respondents: principal components (left), UMAP (centre), and classical multidimensional scaling of the Gower dissimilarity (right). Heights are rescaled so that the modal density equals 100 in each panel, making the shapes comparable while the absolute densities, which depend on the arbitrary scale of each embedding, are not. The PCA and MDS surfaces show a single broad peak; the UMAP surface shows the secondary ridges its construction tends to produce.

UMAP and PCA recover the same dominant axis The correlation between PC1 and the second UMAP coordinate is 91%, and that between PC2 and the first UMAP coordinate is -79%. The principal source of variation in the data — the secular–traditional cleavage — is therefore recovered consistently whether or not linearity is imposed (Figure 4.3). The items most strongly associated with each axis are listed in Appendix A.3.1 (Tables A.4 and A.5).



(a) UMAP embedding versus PCA scores.



(b) Cluster densities along PC1 and PC2.

Figure 4.3: Comparison of the two low-dimensional representations. Panel (a): the UMAP embedding coloured by the k-means partition at $k = 2$ (left) and by PCA score (PC1 centre, PC2 right); the coherent colour gradient across the embedding visually confirms the alignment between the two representations. Panel (b): clusters are ordered and coloured by increasing mean PC1 score, so that colours are comparable across the two algorithms; overlap on PC2 indicates whether UMAP recovers structure beyond the first principal component.

Figure 4.3b illustrates how the clusters are distributed along the previously identified principal component, and compares this distribution with that of the PCA-based clusters along the same component.

4.1.2 A density-based algorithm leaves most respondents unassigned

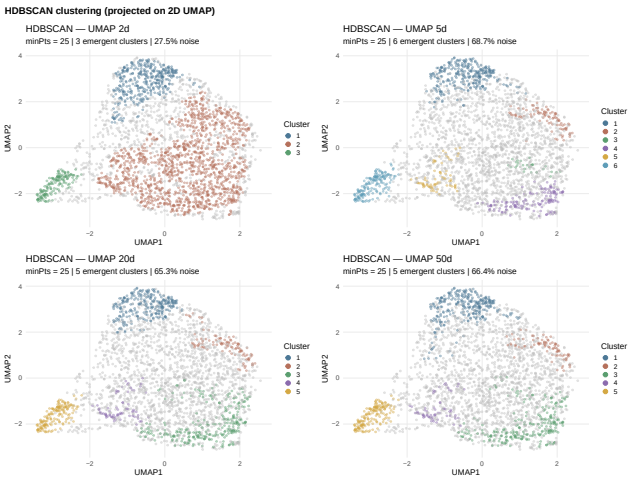


Figure 4.4: HDBSCAN clusterings ($\texttt{minPts} = 25$) projected onto the two-dimensional UMAP for display, by the dimensionality of the embedding actually clustered. Grey points are noise.

Figure 4.4 shows the result. Only a few dense pockets are retained in each case, mostly along the lower-left arm of the cloud, and they occupy the periphery rather than the centre. The centre of the cloud holds the largest number of respondents and is assigned to no cluster at all.

HDBSCAN, which is permitted to leave points unassigned, leaves the center of the cloud unassigned. On the two-dimensional UMAP embedding it recovers three clusters, leaving 27.5% of respondents unassigned. On the five-dimensional UMAP embedding at $\texttt{minPts} = 25$ it recovers six clusters and labels $1{,}771$ of $2{,}579$ respondents — 68.7% — as noise. At twenty and fifty dimensions the noise share is 65.3% and 66.4% (Table A.8 in Appendix A.4). Roughly two respondents in three lie in no region dense enough to constitute a mode. Table 4.1 summarises the two- and five-dimensional runs.

Table 4.1: HDBSCAN on the two- and five-dimensional UMAP embeddings ($\texttt{minPts} = 25$). Agreement with the corresponding k-means partition is computed after Hungarian alignment on the clustered subset only, since noise points have no k-means counterpart to agree or disagree with. The two-dimensional embedding assigns more respondents but, as Table A.9 shows, disagrees with every other dimensionality.

Embedding	Clusters k	Noise (%)	Clustered n	Agreement (%)
UMAP 2D	3	27.5	1,869	71.5
UMAP 5D	6	68.7	808	95.8

Where the algorithm does commit, it commits to the same partition as k-means. On the clustered subset in five dimensions, agreement with KM-UMAP-5D after Hungarian alignment is 95.8%. This is not evidence that the partition is good. The agreement is computed on the 31.3% of respondents HDBSCAN is willing to assign — the densest cores of the embedding — where any two methods would agree, since both are describing the same handful of well-separated nuclei. The figure carrying the substantive point is the noise share, not the agreement conditional on a nucleus existing. Figure 4.5 shows the same comparison in two dimensions, where HDBSCAN assigns more respondents and agreement falls to 71.5%.

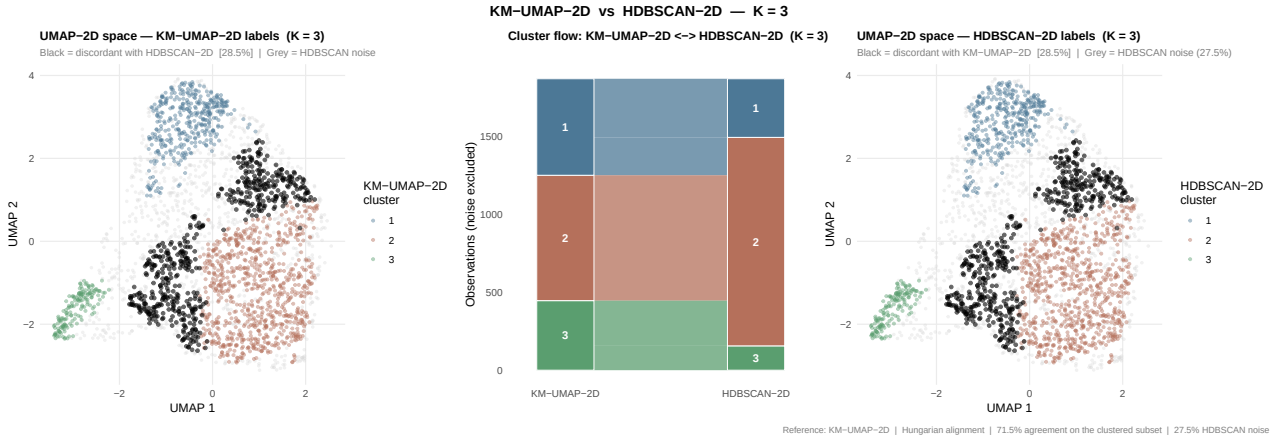
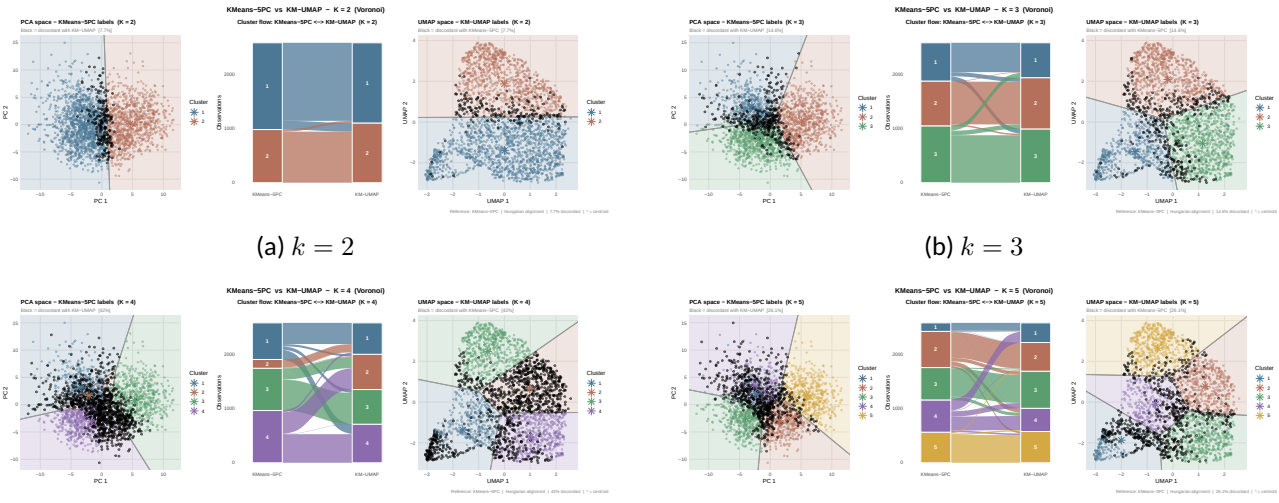


Figure 4.5: k-means and HDBSCAN on the two-dimensional UMAP embedding, $k = 3$. Left and right panels show the two labelings in the same space; grey marks the 27.5% of respondents HDBSCAN leaves unassigned, black the 28.5% of assigned respondents on which the two partitions disagree after Hungarian alignment. The central flow diagram traces respondents between the two labelings, noise excluded. The corresponding figures for the five-dimensional embedding, where HDBSCAN assigns only 31.3% of respondents, are in Table 4.1.

4.1.3 Disagreement between methods concentrates at the boundaries

The representations agree about the centre of each cluster and disagree about its edges. After Hungarian alignment, discordance between KM-PC5 and KM-UMAP is 7.7% at $k = 2$ and 14.6% at $k = 3$, and Figure 4.6 shows the discordant respondents lying almost entirely on the boundaries rather than scattered throughout.



(a) $k = 2$ (b) $k = 3$

(c) $k = 4$ (d) $k = 5$

Figure 4.6: The two embeddings compared at each number of clusters. In each panel the left and right plots show the same respondents labelled by the two methods, with black marking those assigned differently after Hungarian alignment; the centre plot traces respondents between the two labelings. Discordant respondents lie on the boundaries between clusters, not scattered throughout, which is what a continuum predicts: respondents near a dividing line have no determinate membership. Discordance rises from 7.7% at $k = 2$ to 42% at $k = 4$.

This pattern is what a continuum predicts. Where respondents are spread continuously, those near a dividing line have no determinate membership, and different methods place them differently while agreeing about everyone else. Discordance also fails to grow monotonically with k — it reaches 42% at $k = 4$ and falls back to 26.1% at $k = 5$ — which is itself a sign that the boundaries track the algorithm rather than the data. Table 4.2 reports the agreement rate for every pair of configurations.

Table 4.2: Pairwise agreement between clustering configurations, by number of clusters. Entries are the proportion of respondents assigned the same label after Hungarian alignment. Agreement is near-total at $k = 2$ between the two principal-component configurations and degrades sharply from $k = 4$ onwards, most severely against PAM, which works in the native mixed-type space.

Method A	Method B	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
2pc	5pc	99.1%	91.8%	62.3%	71.5%	62.7%
UMAP	5pc	92.3%	85.4%	58.0%	73.9%	66.0%
2pc	PAM	86.6%	68.4%	64.1%	51.8%	54.6%
5pc	PAM	86.7%	68.5%	55.7%	53.8%	45.6%

4.1.4 No selection criterion identifies a number of groups

The silhouette is maximised at $k = 2$, the smallest value examined, and decreases monotonically thereafter (Table A.6 in Appendix A.3.2). A criterion attaining its maximum at the boundary of the grid has not located an optimum; it has failed to find one. The same shape would arise from data with no group structure at all, since two clusters always separate a cloud better than three do when there is nothing to separate. Read together with the elbow, which is likewise monotone, the diagnostics say that no interior optimum exists rather than that the optimum is two.

No partition is stable against individual defection either. Under the fractional hedonic reading of Section 3.4, no partition produced by any method at any k is a Nash equilibrium: in every case some respondents would raise their utility by moving. Groups that do not hold together from their members' point of view are not communities in the sense the substantive question requires. Table A.10 in Appendix A.5.2 reports, for every configuration and every k, the number of respondents with a profitable deviation.

4.1.5 The evidence converges

Table 4.3 collects the diagnostics reported in this section.

Table 4.3: Convergent evidence against a discrete group structure. Each row states what would be observed if American value orientations formed distinct groups, and what is observed instead. The diagnostics are independent: they rest on different assumptions and are computed from different objects.

Diagnostic	Under discrete groups	Observed	Section
Pairwise distances	Bimodal: within- and between-group regimes	Unimodal in all three representations; Gower median 0.581	4.1.1
Density surface	Multiple separated peaks	One broad peak in PCA and MDS	4.1.1
HDBSCAN noise share	Low: most points in a mode	65–69% unassigned for $d \geq 5$	4.1.2
Membership probability	Concentrated near one	98–99.6% of respondents below 0.5	A.4.2
Silhouette width	Local maximum at true k	Maximised at $k = 2$	4.1.4
Nash stability (FHG)	At least one stable partition	None at any k, under any method	4.1.4

The UMAP embedding manufactures apparent clusters from continuous data, for the reasons set out in Section 3.2.2, and the density-based algorithm was run on it rather than on the raw space. The method was given every opportunity to find structure. It found dense pockets at the periphery of a single connected cloud and declined to classify the two thirds of respondents occupying its centre.

4.2 Describing the clusters

4.2.1 Choosing k

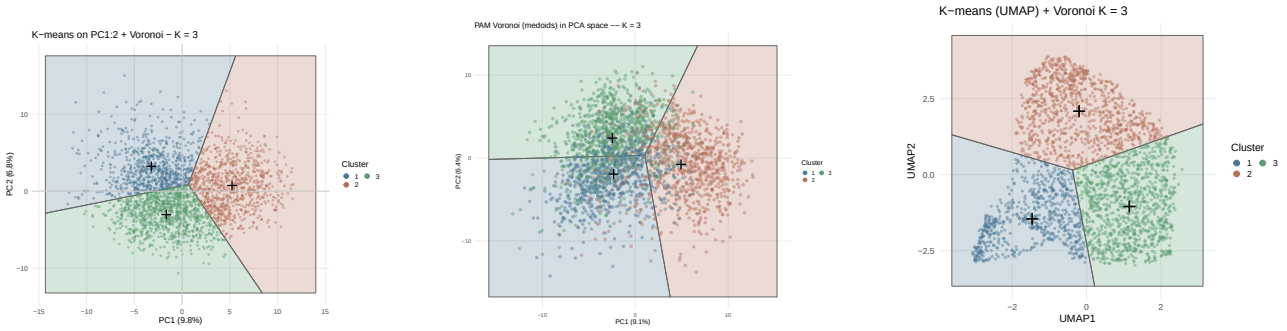
Since no criterion identifies a number of groups, the number is chosen in accordance with the following criteria: robustness across methodologies and differentiation of the information that each cluster recovers.

Our criterion differs from that of the reference study, and the difference follows from the question. Desmet et al. [2025] use Φ_{ST} to measure what share of the variance a partition captures, and are led to $k = 2$. Our object is not variance explained but whether recognisable social identities exist, so the criterion concerns whether each additional group adds interpretable and method-robust content. Both criteria are defensible; neither recovers a discontinuity, because there is none to recover.

We retain $k = 3$: it is the largest partition for which cross-method coherence holds. Beyond three, that coherence breaks, and further splits do not add interpretable information. Cluster sizes under every configuration and every k are reported in Table A.3 (Appendix A.3); Figure 4.7 shows the three-cluster partition under three of the configurations.

4.2.2 Characterizing the clusters

The profiles below rest only on features that survive methodological variation. For each of the four clustering configurations (KM-PC2, KM-PC5, KM-UMAP and PAM-Gower) a one-vs-rest LASSO logistic regression was fitted per cluster, and the selected regressors were refitted without penalty to obtain odds ratios. The core of each profile is made of the variables selected by at least three of the four methods; variables selected by two methods are used only as supporting detail, and those selected by a single method are discarded — roughly three in five of the selected regressors. The reported effect is the geometric mean of the odds ratios across the methods selecting the variable. What remains is what the data say regardless of how the partition is drawn. Table A.7 in Appendix A.3.3 lists these variables for each cluster, with the methods selecting them and the mean odds ratio.



(a) PCA — 2 PCs (b) PAM (c) UMAP

Figure 4.7: The three-cluster partition under each configuration: k-means on two principal components (left), PAM on the Gower dissimilarity shown in principal-component space (centre), and k-means on the UMAP embedding (right). Lines are the Voronoi boundaries between representatives. The point clouds are continuous in all three panels, and the boundaries pass through dense regions rather than through gaps — which is what dividing a continuum requires.

4.2.3 Cluster identities

The first cluster is defined by material precarity and generalised distrust. Its members are more likely to report having gone without food or secure shelter in the past year, and to describe their neighbourhoods as exposed to street violence and to police interference in private life. They express less confidence in the police, in elections, in the press and in international bodies alike, and they say that most people cannot be trusted — including people they know personally and people of other nationalities. Politics matters little to them. They tend to

believe in God, but they do not name religious faith as a quality children should learn, and they are no more likely than others to reject unconventional neighbours. This is not a traditionalist profile. It is a disaffected one: belief without institutional attachment, and disadvantage without ideological expression.

The second cluster is defined by secularism combined with institutional cosmopolitanism. Its members reject belief in God, Hell and Heaven, refuse the claim that religion prevails over science, and describe themselves as non-religious or atheist. They are materially secure. They disagree that immigration raises the risk of terrorism, favour more open immigration policy, and hold that immigration improves the lives of people from poorer countries. They report higher confidence in the World Health Organization, in NATO and in the World Trade Organization. They prioritise free speech over the maintenance of order, favour environmental protection over economic growth, reject the idea that men make better political leaders, and value imagination as a quality to be cultivated in children. Their distinguishing trait is not the absence of trust but its redirection: away from religious authority and towards supranational institutions.

The third cluster is defined by religious practice coupled with confidence in domestic authority. Its members believe in God, Hell and Heaven, identify as religious, and express confidence in the churches. They also express confidence in the government, in the police and in major companies, perceive less corruption in the civil service, and report calm neighbourhoods and little personal experience of crime. They treat greater respect for authority as desirable, name obedience as a quality children should learn, and select the maintenance of order and the fight against crime as national priorities. They agree that immigration increases the risk of terrorism. Order is the organising principle here, and religious and civil authority are not opposed but continuous with one another.

4.3 Recovering cluster membership

4.3.1 What the exercise measures

In this second stage of the analysis, by building a classifier to recover cluster membership, we look into the relationship between values and sociodemographic characteristics on the one hand, and between values and party affiliation on the other. This section asks the question that survives that answer: if value orientations are not divided into groups, is a person's position along the gradient at least fixed by the identities we can observe? Party affiliation is the most visible of those identities, and social condition is the one identity economics ordinarily builds groups on. The comparison between the two is the point of what follows.

What a classifier measures is therefore not the accuracy of the clustering but the *recoverability* of its labels from information the clustering never saw — external separability, not validation.

For each partition at $k = 3$ a multinomial LASSO logistic regression and a random forest are fitted on a training split and evaluated out of sample. Three predictor blocks are compared: the 60 political variables, the 27 sociodemographic variables, and their union. Two quantities are reported. Balanced accuracy averages recall across clusters, so its chance level is $1/3$ whatever the cluster sizes. Cohen's κ is computed against the agreement expected from the empirical cluster distribution (κ_{emp}), which is the relevant correction because the clusters are unequal in size and raw accuracy rewards predicting the largest one.

4.3.2 Politics recovers more than social condition

Table 4.4: Out-of-sample recovery of cluster membership from the held-out blocks, $k = 3$, full sample ($n = 2{,}579$, 5-fold cross-validation). Balanced accuracy averages recall over the three clusters, so its chance level is 0.333; κ_{emp} corrects observed agreement for the empirical cluster distribution. The last column names the cluster with the lowest recall under the LASSO, with that recall in parentheses, using the labels of Section 4.2. The two classifiers agree on the least-recovered cluster in every cell except KM-PC5 under the sociodemographic block, where the disaffected and secular clusters are recovered within 0.002 of each other.

		Balanced accuracy		κ_{emp}		
Block	Partition	LASSO	RF	LASSO	RF	Least recovered
Political	KM-PC2	0.688	0.681	0.532	0.524	Disaffected (0.553)
	KM-PC5	**0.700**	**0.702**	**0.556**	**0.560**	Disaffected (0.565)
	KM-UMAP	0.682	0.688	0.538	0.545	Disaffected (0.541)
	PAM-Gower	0.603	0.622	0.404	0.433	Disaffected (0.500)
Sociodemographic	KM-PC2	0.600	0.597	0.401	0.393	Secular (0.557)
	KM-PC5	**0.610**	**0.613**	**0.416**	**0.419**	Disaffected (0.556)
	KM-UMAP	0.596	0.594	0.396	0.392	Disaffected (0.523)
	PAM-Gower	0.538	0.525	0.306	0.288	Disaffected (0.509)
Both	KM-PC2	0.713	0.706	0.569	0.561	Disaffected (0.601)
	KM-PC5	**0.725**	**0.724**	**0.591**	**0.594**	Disaffected (0.620)
	KM-UMAP	0.713	0.703	0.580	0.567	Disaffected (0.605)
	PAM-Gower	0.624	0.625	0.437	0.437	Disaffected (0.514)

Three readings follow from Table 4.4. The first is the one the section exists for: political attitudes recover the partition about a third better than social condition does, under every partition and both classifiers, and adding demography to politics buys between 0.033 and 0.042 points of κ_{emp}. The second is that the LASSO and the random forest agree to within 0.03 everywhere, so nothing here depends on assuming linearity. The third is that recovery is uneven in a systematic way.

Combining the two blocks adds little. Balanced accuracy rises to 0.624–0.725 and κ_{emp} to 0.437–0.591, a mean gain of roughly $+0.03$ over the better of the two blocks used alone. The blocks are therefore largely redundant: most of what social condition says about a respondent's values has already been said by their politics. Appendix A.6 reports the same exercise for every number of clusters from two to six (Tables A.11–A.14).

The comparison is between the two blocks as the questionnaire constitutes them, not between the concepts they stand for: the political block contains twice as many variables as the sociodemographic one, and a larger block has more chances of containing a useful predictor. The gap survives in the random forest, which is not subject to LASSO's selection, but it is not a measure of how much partisanship matters in the abstract.

4.3.3 Which cluster is recovered

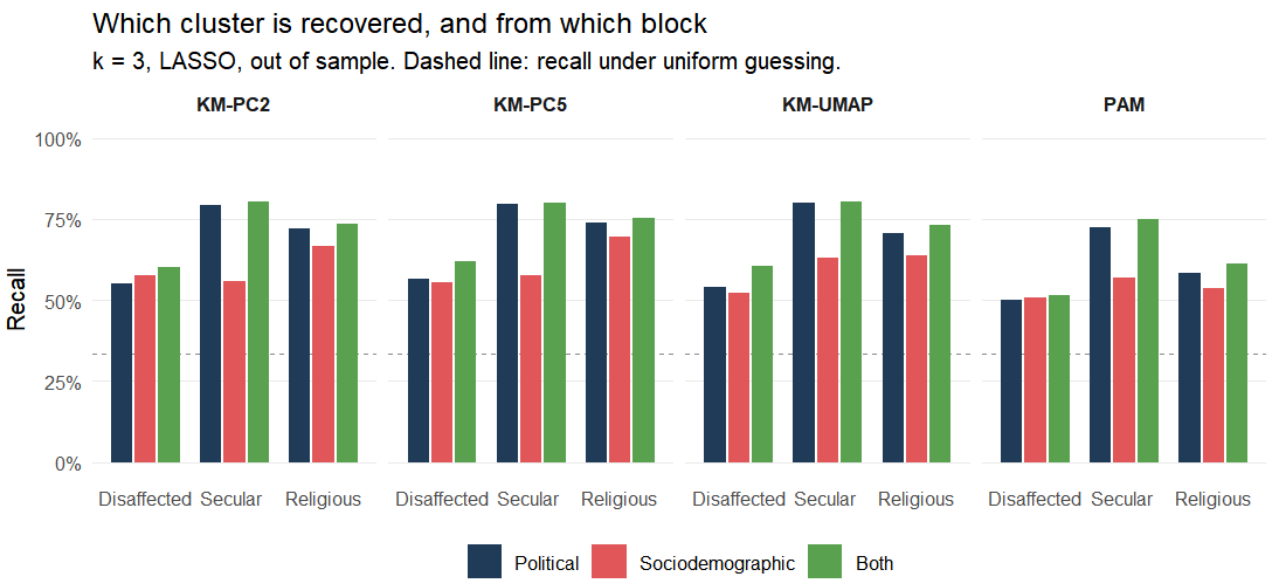


Figure 4.8: Out-of-sample recall of each cluster at $k = 3$, by predictor block and clustering configuration (LASSO, full sample). Bars are the share of the members of a cluster that the classifier places in it; the dashed line is the recall a uniform random assignment would achieve. Cluster labels follow Section 4.2: *Disaffected* (precarity and distrust), *Secular* (secular-cosmopolitan), *Religious* (faith and order). Recall is asymmetric across clusters and the asymmetry is driven by the political block, not by the sociodemographic one.

The disaffected cluster is the least recovered almost everywhere (Figure 4.8; Table 4.5). Under the political block it is identified correctly for between 50 and 57 per cent of its members, against 79 to 80 per cent for the secular cluster.

The secular and religious clusters occupy the two ends of the religious–secular axis, and that axis is the one around which American party competition is organised, so a respondent's position on it can be read off their political attitudes almost directly. The disaffected cluster is not defined by a position on that axis at all: its members are distinguished by deprivation and by withdrawal from institutions, including from politics itself. A block of political variables has little purchase on people whose defining trait is disengagement from politics, and the low recall records that absence rather than a failure of the classifier.

Table 4.5: Per-cluster recall at $k = 3$, full sample ($n = 2{,}579$, 5-fold cross-validation), under LASSO-penalised multinomial regression and the random forest. Recall is the share of a cluster's true members correctly identified; n is the number of true members. Cluster labels are Hungarian-aligned to KM-PC5.

			Disaffected		Secular		Religious	
Block	Partition	$n_1/n_2/n_3$	LASSO	RF	LASSO	RF	LASSO	RF
Political	KM-PC2	771/796/1012	55.3%	54.1%	79.1%	76.5%	72.0%	73.7%
	KM-PC5	705/826/1048	56.5%	56.6%	**79.8%**	77.2%	73.8%	**76.6%**
	KM-UMAP	643/945/991	54.1%	56.3%	**79.9%**	75.4%	70.6%	74.7%
	PAM-Gower	814/847/918	50.0%	53.4%	72.4%	71.1%	58.4%	62.1%
Sociodemographic	KM-PC2	771/796/1012	57.7%	58.1%	55.7%	56.5%	66.6%	64.3%
	KM-PC5	705/826/1048	55.6%	57.2%	57.7%	57.0%	**69.6%**	**69.7%**
	KM-UMAP	643/945/991	52.3%	52.1%	62.9%	62.5%	63.8%	63.7%
	PAM-Gower	814/847/918	50.9%	47.5%	56.9%	56.7%	53.5%	53.4%
Both	KM-PC2	771/796/1012	60.1%	59.1%	80.3%	77.1%	73.6%	75.6%
	KM-PC5	705/826/1048	62.0%	61.0%	80.1%	78.2%	**75.4%**	**78.1%**
	KM-UMAP	643/945/991	60.5%	58.6%	**80.3%**	76.8%	73.1%	75.6%
	PAM-Gower	814/847/918	51.4%	52.7%	74.9%	72.5%	61.1%	62.2%

4.3.4 Which observable characteristics carry the signal

Table 4.6: Which held-out items the two classifiers rely on, $k = 3$. Entries count how many of the four clustering configurations place the item among the twenty strongest terms under each criterion, so the maximum is four. Only items selected by at least three configurations under at least one criterion are shown.

Variable	Description	Random forest	LASSO
Political block			
Q223	Party vote intention	4	4
Q239	System governed by religious law	4	4
Q240	Left–right self-placement	4	4
Q249	Women have the same rights as men	4	4
Q254	National pride	4	4
Q235	Strong leader without parliament	4	4
Q246	Civil rights protect from state oppression	4	4
Q242	Religious authorities interpret the law	4	4
Q210	Joining boycotts	4	–
Q237	Army rule	4	–
Sociodemographic block			
Q289	Religious denomination	3	4
Q290	Ethnic group	3	4
Q287	Subjective social class	3	4
Q275	Education (respondent)	3	4
Q288	Household income decile	4	–

Table 4.6 lists the items the classifiers rely on. What the political block retains is regime preference, not participation. Vote intention, left–right placement and five items from the battery on what democracy requires are selected unanimously. Boycotts, demonstrations and army rule rank highly for the random forest but never survive the LASSO penalty. The gradient tracks what people think the political system should be, not how much they take part in it.

What the sociodemographic block retains are the characteristics on which identity economics actually builds identities: ethnicity, subjective class, education, and religious denomination. That these are precisely the variables selected, unanimously under the LASSO, is itself informative: it draws a direct link between where a respondent sits on the value gradient and who they take themselves to be, not what they materially possess. Household income ranks for the random forest but is never selected once subjective class is available, and the

distinction matters for reading the first cluster: its members are (or at least declare to be) materially precarious, but it is not material position that locates them on the axis.

Both lists must be read at the level of the block rather than the item. The LASSO selects arbitrarily among collinear predictors, and the democracy battery alone contains nine closely related items, so which of them is retained is not stable across resamples.

4.3.5 The density-based partition

HDBSCAN offers a check of a different kind, because it is free to leave respondents unassigned. Its labels can be put through the same exercise, but only on the $1{,}869$ respondents it does assign. Those are the densest regions of the embedding, where any classifier should do better, so comparing them with the full-sample figures would reward the selection rather than the partition. Table 4.7 avoids this by refitting the other four configurations on exactly the respondents HDBSCAN retains.

Table 4.7: Recovery at $k = 3$ on the $1{,}869$ respondents HDBSCAN assigns to a core, corresponding to 72.5% of the sample; the remaining 710 respondents are left unassigned. All five partitions are estimated on the same respondents, making the results directly comparable. Although HDBSCAN attains the highest overall accuracy in all three blocks, its lower balanced accuracy indicates that this result is partly driven by cluster-size imbalance.

	Political			Sociodemographic			Both		
Configuration	Acc.	Bal. acc.	κ_{emp}	Acc.	Bal. acc.	κ_{emp}	Acc.	Bal. acc.	κ_{emp}
HDBSCAN	**81.9%**	61.9%	0.541	**77.7%**	50.4%	0.386	**85.4%**	65.2%	**0.632**
KM-PC2	70.0%	68.4%	0.534	61.8%	**60.3%**	0.406	72.4%	71.3%	0.574
KM-PC5	72.2%	70.3%	0.566	62.7%	60.0%	**0.409**	74.2%	**72.4%**	0.597
KM-UMAP	72.7%	**70.4%**	**0.573**	61.9%	59.5%	0.400	73.8%	72.0%	0.593
PAM-Gower	61.4%	60.8%	0.415	54.6%	53.7%	0.310	61.3%	60.7%	0.414

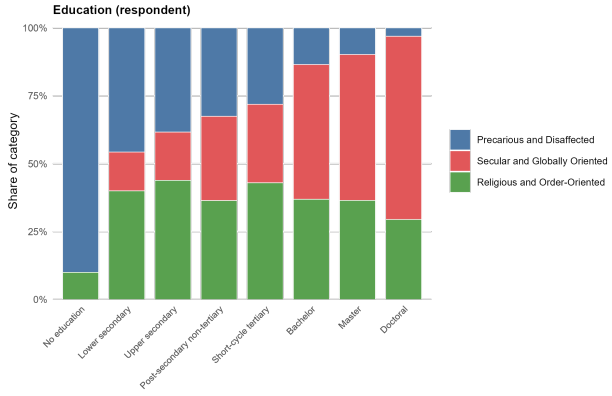
This is the clearest illustration in the thesis of why accuracy is the wrong measure here. Under the political block HDBSCAN is recovered nine points more accurately than the best k-means configuration and worse than two of them once chance agreement is removed. The reason is in the shape of the partition rather than in the data: one core holds $1{,}337$ of the $1{,}869$ assigned respondents, so predicting it for everyone is already right 71.5% of the time.

Recoverability improves when the ambiguous cases are removed, which is what one expects of a continuum and not a demonstration that the cores are real groups. Table 4.8 shows how unevenly the three cores are recovered.

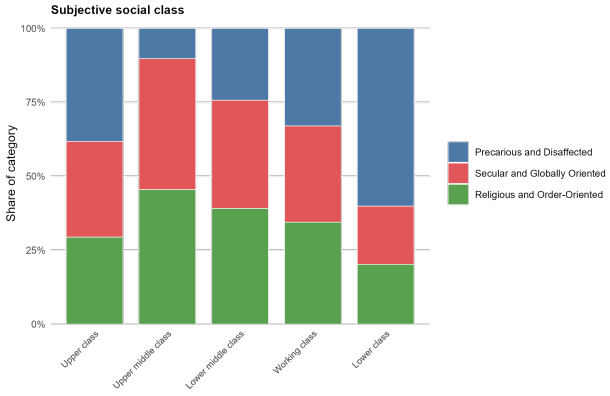
Table 4.8: Out-of-sample recall and precision for each HDBSCAN core, LASSO, by predictor block. Support is the number of respondents assigned to the core. The three cores are strongly unequal: one holds 72% of the assigned respondents, and the smallest holds 8%.

		Political		Sociodemographic		Both	
Core	Support	Recall	Prec.	Recall	Prec.	Recall	Prec.
1	376	0.660	0.763	0.476	0.713	0.761	0.864
2	1,337	0.927	0.837	0.941	0.791	0.952	0.860
3	156	0.269	0.677	0.096	0.536	0.244	0.667

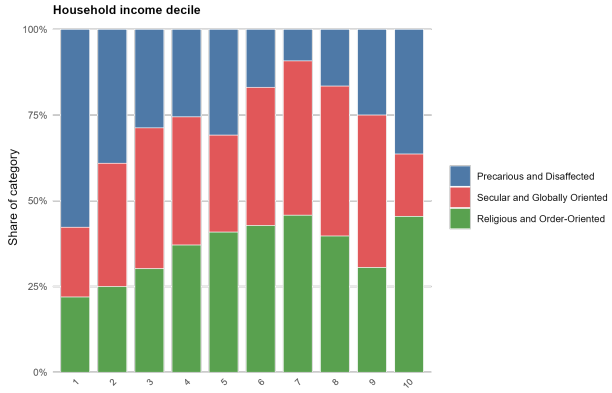
4.3.6 Observable characteristics tilt a position without fixing it



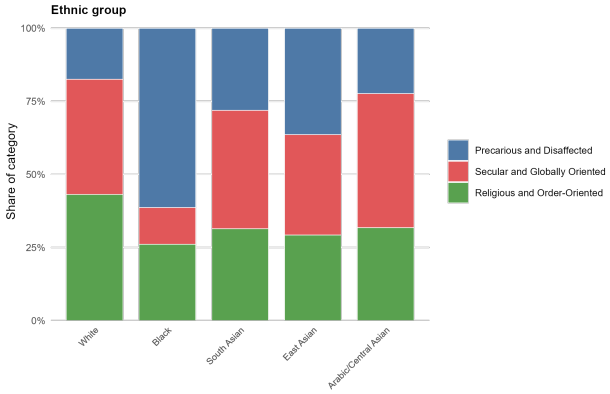
(a) Education



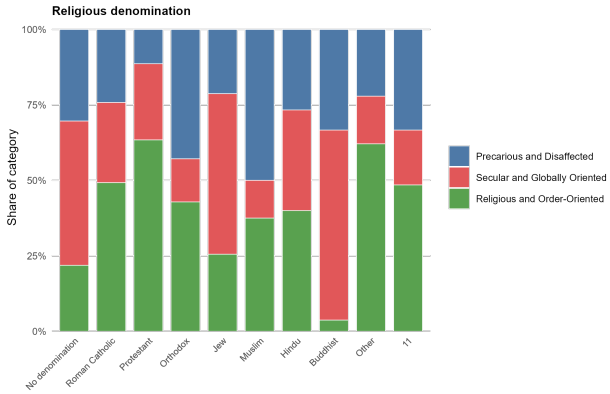
(b) Subjective social class



(c) Household income decile



(d) Ethnic group



(e) Religious denomination

Figure 4.9: Composition of each sociodemographic category by cluster, $k = 3$, KM-PC5. Each bar is one category of the variable named in the panel and sums to one hundred per cent; the coloured segments give the share of that category's members falling in each cluster. A category dominated by a single colour would be a category whose members share a value position. No category in any panel is dominated by a single colour: every one of them contains a substantial share of all three clusters. The panels are drawn from the five sociodemographic items the classifiers rely on most (Table 4.6).

Figure 4.9 shows the composition of each category by cluster. Education moves the composition monotonically: the disaffected cluster falls from 89% among respondents with no formal education to 4% among doctoral holders, and the secular cluster rises across the same range. Subjective social class moves it in the same direction

and more sharply at the lower end, with the disaffected share highest among those describing themselves as lower class.

Income behaves differently: it separates the disaffected cluster from the other two at the bottom of the distribution and says almost nothing above the median, where the three shares are roughly flat. This is consistent with a cluster defined by deprivation rather than by a place on the value axis: what matters is whether a respondent has gone without, not how much they have. It also explains why household income never survives the LASSO once subjective class is available, as reported in Section 4.3.4: the two carry the same information about the bottom of the distribution, and subjective class carries more about the rest.

Denomination is the strongest single predictor in the panel and still does not sort people. Respondents reporting no denomination are the most concentrated group in the figure, but the remaining are distributed across the other two. Religious affiliation moves a respondent's expected position substantially and fixes it for nobody.

This is the answer to the second question of Section 1.2. The three partitions do not coincide. The sociodemographic categories on which identity economics ordinarily builds groups do not determine a value position: they shift the odds, weakly, and largely through partisanship, which the previous sections showed carries most of what they have to say.

Politics comes closest, and only on the axis it is itself organised around; social condition adds almost nothing politics has not already said; and neither approaches what a genuinely discrete partition would allow.

Observable categories are not irrelevant to values: they shift the odds, and in the case of denomination they shift them a great deal. What they do not do is determine a position, and the practice of treating them as a proxy for value identity assumes a correspondence the data do not support. A group defined by observable characteristics and a group defined by shared values are different objects. Building the second out of the first is not a shortcut but a substitution, and Chapter 5 takes up what follows for how the question should be posed instead.

5. Conclusion

This thesis began from an idea borrowed from economics: people belong to groups, and the group they belong to shapes what they do. To use that idea one has to know which groups exist. The usual answer builds them from what can be seen from the outside, such as income, education, race or party. A more recent answer builds them from what people believe, placing together those who give similar answers to questions about religion, family, trust, money and right and wrong. This second answer is attractive because it defines a group by what its members have in common rather than by a label assigned to them. It also rests on an assumption that is rarely tested: that beliefs sort people into groups in the first place. This thesis tested it.

The material was the American sample of the World Values Survey, 2,579 people answering 198 questions about what they value. Questions about politics and personal background were deliberately set aside, so that any group found could not be a restatement of party or of income. The same people were then described in three different ways and searched for groups with four different methods, including one that is allowed to answer that there are none and one that is known to produce groups even when there is nothing to find. The point of the redundancy was to make a negative answer hard to blame on the tool.

The answer is negative. People are not close to some of their compatriots and far from the rest; the distances between them form one smooth spread with no gap in it. The population has one crowded middle rather than several separate concentrations. The method free to find nothing left two respondents in three unclassified, and placed the small dense pockets it did find at the edges of the crowd, never at its centre. Finally, no division of the sample was one its own members would accept: under every method and every number of groups, a substantial share of people would rather be somewhere else than where they were put. Americans are spread out on values. They are not sorted into camps.

Three positions were nevertheless retained, and the choice was ours rather than the data's. Three is simply the largest number of positions that keeps its meaning when the method changes. The first is a position of hardship and mistrust: people who have gone without food or shelter, who see violence around them, who trust neither institutions nor their own neighbours, and who take little interest in politics. The second is secular and outward-looking: people who reject religious belief, feel materially secure, and place their confidence in international institutions rather than churches. The third joins religious practice to confidence in domestic authority, in the police, the government and the churches, and puts order first. These are regions of a gradient. Most people live between them, not in them.

Observable characteristics tilt a person's position without fixing it. Knowing someone's political attitudes places them correctly about seven times in ten, roughly a third better than knowing their income, age, education and employment. Adding social background to politics adds almost nothing, which means the two say largely the same thing. Nothing comes close to the accuracy that real, separate groups would allow, and the same conclusion is visible without any model at all: every age group and every income bracket contains a substantial share of all three positions. The categories on which identity economics usually builds groups do not determine what a person values. They lean on it, weakly, and mostly through politics.

There is a plausible reason why the search failed, and it points to what should be done next. Likeness between people is real, but it is local. Religious belief travels with views on sexuality; economic security travels with confidence in institutions. What does not exist is likeness across everything at once. Measuring how similar two people are by averaging 198 questions asks whether they resemble each other in general, and almost nobody resembles anybody in general. A group may well be a group on one subject and not on another, and a person may belong to different groups depending on the subject at hand. The natural next step is therefore to look for groups inside each theme separately, and then ask whether a person's group on religion is the same as their group on the economy. If it is not, then the question "which group does this person belong to" has no single answer, and the more useful question becomes which group governs which decision.

Two limits should be kept in mind. The evidence comes from one country in one year, so nothing here says whether the continuum is American or general; repeating the exercise on other national samples would settle it cheaply. And the comparison between politics and social background is a comparison between two sets of survey questions, not between two ideas: the survey asks twice as much about politics as about circumstances, and a longer list has more chances of containing something useful.

Bibliography

George A. Akerlof and Rachel E. Kranton. Economics and identity. *The Quarterly Journal of Economics*, 115(3): 715–753, 2000. doi: 10.1162/003355300554881.

Marianne Bertrand and Emir Kamenica. Coming apart? cultural distances in the united states over time. NBER Working Paper 24771, National Bureau of Economic Research, Cambridge, MA, 2018.

Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20 (1):37–46, 1960. doi: 10.1177/001316446002000104.

Klaus Desmet, Ignacio Ortuño-Ortín, and Romain Wacziarg. Latent polarization. NBER Working Paper 34229, National Bureau of Economic Research, Cambridge, MA, September 2025. Also circulated as CEPR Discussion Paper 20644; earlier version titled "Endogenous Partitions".

Christian Haerpfer, Ronald Inglehart, Alejandro Moreno, Christian Welzel, Kseniya Kizilova, Jaime Diez-Medrano, Marta Lagos, Pippa Norris, Eduard Ponarin, and Bi Puranen. World values survey: Round seven — country-pooled datafile version 5.0. JD Systems Institute and WVSA Secretariat, Madrid and Vienna, 2022. United States sample, `USA_2017_WVS-W7_v01_M`.

Ronald Inglehart and Christian Welzel. *Modernization, Cultural Change, and Democracy: The Human Development Sequence*. Cambridge University Press, Cambridge, 2005.

Mehmet Karakaya. Hedonic coalition formation games: A new stability notion. *Mathematical Social Sciences*, 61(3):157–165, 2011. doi: 10.1016/j.mathsocsci.2011.03.004.

National Center for Education Statistics. *Digest of Education Statistics 2020*. U.S. Department of Education, Washington, DC, 2021. Table 101.20, resident population by race/ethnicity.

A. Appendix

The appendices collect the structure of the questionnaire, the treatment of non-response, supplementary results for the clustering stage (including the variables on which the cluster profiles rest), the density-based robustness check, the coalition-theoretic treatment of partition stability, and the recovery results for every number of clusters and every predictor block.

A.1 Questionnaire Structure

Thematic Sections of the Questionnaire The questionnaire is organised in fourteen thematic sections. Only the eleven listed under *Latent variables* enter the clustering stage; the remaining three are held out until the recovery stage.

Table A.1: WVS Wave 7 — thematic sections, grouped by variable block

Section	Q range	#	Description
Latent variables			
Social Values, Attitudes & Stereotypes	Q1–Q45	45	Life domains; child qualities; neighbour tolerance; gender attitudes
Happiness & Well-Being	Q46–Q56	11	Happiness; health; life & financial satisfaction; deprivation
Social Capital, Trust & Membership	Q57–Q105	49	Generalised & interpersonal trust; institutional confidence; org. membership
Economic Values	Q106–Q111	6	Equality vs. incentives; ownership; govt. responsibility; competition
Corruption	Q112–Q120	9	Perceived corruption; personal experience; accountability risk
Migration	Q121–Q130	10	Immigration impact; specific effects; preferred policy
Security	Q131–Q150	20	Personal security; neighbourhood disorder; crime; geopolitical threats
Postmaterialist Index	Q151–Q157	7	Willingness to fight; societal goal priorities (Inglehart battery)
Science & Technology	Q158–Q163	6	Attitudes toward S&T; perceived societal effects
Religious Values	Q164–Q175	12	Importance of God; beliefs; practice frequency; self-identification
Ethical Values & Norms	Q176–Q198	23	Moral justifiability (10-pt scales); surveillance rights
Political variables			
Political Interest & Participation	Q199–Q234	36	Political interest; media use; political actions; voting; election quality
Political Culture & Regimes	Q235–Q259	25	Regime preferences; left–right placement; democratic values; national pride
Sociodemographic variables			
Demographics	Q260–Q290	31	Sex; age; education; employment; income; religion; ethnicity

A.2 Missing Value Treatment

The WVS has no native `NA` value: non-response is encoded through four reserved negative integers — -1 *don't know*, -2 *no answer or refused*, -3 *not applicable* (questionnaire filter), and -5 *missing for other reasons*. All four are recoded to `NA` before any further processing, following Desmet et al. [2025]; Section A.2.2 discusses the information this discards.

A.2.1 Filtering and Imputation

Response-rate thresholds Two filters are applied sequentially to the recoded matrix. Any variable answered by fewer than 30% of respondents is removed first; any respondent answering fewer than 70% of the *surviving* variables is removed second. The order matters: respondents are not penalised for skipping items already discarded as uninformative. Applied to the U.S. sample the filters retain $n = 2{,}579$ of the $2{,}596$ original respondents. The surviving row indices are stored and reused for every downstream object — imputed data, cluster labels, geographic covariates — so that positional alignment is preserved without an explicit join key.

Random Forest imputation Residual missingness is imputed with a Random Forest classifier. All variables are treated as factors throughout, so that imputation is a classification rather than a regression problem and no artificial metric is imposed on categorical scales. The subset of respondents with no missing value on any retained variable forms the training set T, with $|T| = 253$; the remaining $2{,}326$ respondents form the imputation target. Because the forest requires complete predictor vectors at prediction time, missing entries are first filled with the modal category of their variable; this initialisation affects only the predictors, never the imputed value itself. For each variable with residual missingness a forest is then fitted on T using all other retained variables as predictors — 500 trees, $\lfloor\sqrt{p}\rfloor$ candidate predictors per split, minimum node size 10 — and applied to the rows where that variable was originally missing. The procedure is run once rather than iteratively: imputed values are not fed back as predictors beyond the modal initialisation. After imputation the working matrix contains no missing entries.

A.2.2 Limitations

Size of the training set The imputation model is fitted on 253 complete respondents and used to impute the remaining 90% of the sample. A training set this small relative to the target limits the variance the forest can capture, and imputed values tend to regress toward the response patterns of the complete-case subpopulation. Complete responders are not a random subsample — by construction they are the respondents who answered every sensitive item — so the bias is conservative, pulling imputed profiles toward the centre of the distribution. This works against finding discrete structure only in the weak sense of smoothing the data: it does not manufacture the continuum reported in Section 4.1, which is also visible in the complete-case subsample alone.

Informative missingness Recoding all four reserved codes to a single `NA` discards the information carried by the *type* of non-response (Table A.2). Two patterns matter. Refusals (-2) concentrate on a few sensitive or knowledge-based items — Q220 (organising political activities via the internet, 465 refusals), Q92 and Q91 (factual questions on the IMF and the UN Security Council, 414 and 410). On knowledge items a refusal plausibly encodes absence of knowledge rather than unwillingness to answer, so imputing a substantive category assigns respondents an answer they did not have. Not-applicable codes (-3) concentrate on four occupational and educational items: Q282 (spouse's occupational group, 1,545 codes), Q276 (spouse's education, 1,093), Q280 (spouse's employment status, 1,093) and Q281 (respondent's own occupational group, 855). For the three spouse variables the code is structurally determined — respondents without a partner were never asked — and imputation fabricates spousal characteristics for roughly half the sample. All four belong to the sociodemographic block and so never enter the clustering input; in the recovery stage they are dropped rather than imputed.

Table A.2: Missing values by thematic section before imputation, disaggregated by type of non-response. WVS Wave 7, U.S. sample.

Section	Missing %	% of all missing	Total	Don't know	Refused	Not appl.
Demographics	9.7%	36.8%	6,783	898	1,299	4,586
Social Capital & Trust	2.2%	15.3%	2,814	169	2,645	0
Political Interest	3.0%	14.9%	2,744	54	2,690	0
Social Values	1.6%	10.2%	1,888	915	973	0
Political Culture	1.9%	6.6%	1,225	63	1,162	0
Ethical Values & Norms	1.1%	3.7%	676	14	662	0
Security	0.9%	2.6%	488	21	467	0
Religious Values	1.5%	2.6%	477	15	462	0
Postmaterialist Index	2.3%	2.0%	362	7	355	0
Migration	1.3%	1.9%	342	13	329	0
Corruption	0.9%	1.1%	205	20	185	0
Science & Technology	1.1%	1.0%	177	6	171	0
Happiness & Wellbeing	0.5%	0.7%	135	1	134	0
Economic Values	0.8%	0.7%	132	7	125	0
Total		100%	18,448	2,203	11,659	4,586

A.3 Supplementary Results for the Clustering Stage

Cluster Sizes Table A.3 reports the share of respondents in each cluster, for every configuration and every k from two to six.

Table A.3: Cluster size distribution (%) by method and number of clusters k.

Method	k	$c=1$	$c=2$	$c=3$	$c=4$	$c=5$	$c=6$
KM-PC2	2	61.7	38.3	—	—	—	—
	3	29.9	30.9	39.2	—	—	—
	4	25.7	24.2	21.0	29.0	—	—
	5	22.4	20.9	24.1	14.2	18.5	—
	6	19.9	14.8	20.4	13.7	9.6	21.6
KM-PC5	2	62.0	38.0	—	—	—	—
	3	27.3	32.0	40.6	—	—	—
	4	26.2	6.2	30.2	37.3	—	—
	5	6.1	25.9	23.1	23.0	21.9	—
	6	20.9	18.6	21.1	18.0	5.3	16.1
KM-UMAP	2	57.3	42.7	—	—	—	—
	3	24.9	36.6	38.4	—	—	—
	4	22.5	25.2	24.8	27.6	—	—
	5	14.0	20.5	26.5	16.6	22.3	—
	6	19.0	20.4	18.1	15.3	11.8	15.4
PAM	2	61.4	38.6	—	—	—	—
	3	31.6	32.8	35.6	—	—	—
	4	26.5	21.5	24.0	28.0	—	—
	5	11.6	20.6	27.1	23.2	17.5	—
	6	20.1	16.8	23.3	20.2	8.3	11.3

A.3.1 Interpretation of the Representations

Loadings of the First Two Principal Axes PC1 opposes moral permissiveness to religiosity, with confidence in international bodies loading on the permissive side. Nine of the ten strongest loadings on PC2 come from the

Social Capital & Trust section (Table A.4).

Table A.4: Ten largest absolute loadings on PC1 and PC2, under z-score standardisation

PC1

Variable	Section	Loading
Q182	Ethical Values	$+0.153$
Q169	Religious Values	$+0.146$
Q184	Ethical Values	$+0.141$
Q88	Social Capital & Trust	-0.140
Q36	Social Values	-0.137
Q160	Science & Technology	-0.137
Q79	Social Capital & Trust	-0.134
Q80	Social Capital & Trust	-0.132
Q164	Religious Values	-0.130
Q66	Social Capital & Trust	-0.130

PC2

Variable	Section	Loading
Q64	Social Capital & Trust	$+0.165$
Q69	Social Capital & Trust	$+0.157$
Q70	Social Capital & Trust	$+0.142$
Q65	Social Capital & Trust	$+0.139$
Q77	Social Capital & Trust	$+0.137$
Q76	Social Capital & Trust	$+0.135$
Q78	Social Capital & Trust	$+0.134$
Q171	Religious Values	$+0.128$
Q59	Social Capital & Trust	$+0.126$
Q94	Social Capital & Trust	-0.126

Items Associated with the UMAP Coordinates The UMAP coordinates admit no loadings, since the mapping is not linear, so items are associated with axes by permutation importance in a random forest predicting each coordinate from the value items (Table A.5).

Table A.5: Ten survey items most strongly associated with each UMAP coordinate (Euclidean embedding). %IncMSE is the percentage increase in mean squared error under permutation of the item, averaged over 500 trees.

UMAP1

Variable	Section	%IncMSE
Q189	Ethical Values & Norms	28.22
Q165	Religious Values	26.54
Q64	Social Capital & Trust	24.98
Q171	Religious Values	23.43
Q192	Ethical Values & Norms	22.92
Q179	Ethical Values & Norms	22.18
Q134	Security	21.59
Q69	Social Capital & Trust	21.37
Q51	Happiness & Wellbeing	18.07
Q86	Social Capital & Trust	17.90

UMAP2

Variable	Section	%IncMSE
Q182	Ethical Values & Norms	35.45
Q169	Religious Values	30.18
Q80	Social Capital & Trust	20.54
Q88	Social Capital & Trust	20.50
Q160	Science & Technology	18.42
Q130	Migration	18.22
Q126	Migration	17.46
Q186	Ethical Values & Norms	17.41
Q161	Science & Technology	17.18
Q164	Religious Values	16.94

A.3.2 Structure Diagnostics

Table A.6: Average silhouette width by clustering method and number of clusters.

Method	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
K-means (2 PCs)	**0.0592**	0.0476	0.0291	0.0271	0.0162
K-means (5 PCs)	0.0590	0.0486	0.0473	0.0308	0.0303
K-means (UMAP)	0.0563	0.0480	0.0284	0.0271	0.0200
PAM	0.0561	0.0389	0.0280	0.0188	0.0194

Silhouette Score Table A.6 reports the average silhouette width for each clustering method and value of k. All methods achieve their highest score at $k = 2$, although the generally low values indicate weakly separated and potentially overlapping clusters. Among the evaluated solutions, K-means using two principal components at $k = 2$ yields the highest silhouette width.

Label Alignment and Measures of Association Cluster labels are arbitrary, so the agreement rates of Section 4.1.3 are computed after aligning labels. The alignment maximises the trace of the contingency table over label permutations,

$$\boldsymbol{\pi}^* = \arg\max_{\boldsymbol{\pi} \in \mathcal{S}_k} \sum_{j=1}^{k} M_{j,\boldsymbol{\pi}(j)}, \tag{A.1}$$

where M is the $k \times k$ contingency table between the two partitions and $\mathcal{S}_k$ the set of permutations of $\{1, \ldots, k\}$. The maximum is found by the Hungarian algorithm, which is exact for any k and does not rely on ordering the clusters along any axis. The reference partition is left unchanged and the other relabelled.

Definition of the Silhouette Width Let $a(i)$ be the average dissimilarity between respondent i and the other members of its own cluster, and $b(i)$ the smallest average dissimilarity between i and the members of any other cluster. The silhouette width of i is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i),\, b(i)\}} \in [-1, 1], \tag{A.2}$$

and the criterion used in Section 3.4 is its average over respondents. Values near one indicate a respondent well inside its cluster, values near zero one sitting on a boundary, and negative values one closer to another cluster than to its own. Dissimilarities are Euclidean for the k-means configurations and Gower for PAM, matching the metric each configuration was fitted with.

Two properties bear on how the criterion is read. Unlike the within-cluster sum of squares, which decreases mechanically in k and therefore has no interior optimum, the silhouette can have one, because it rewards separation from the nearest alternative as well as cohesion. Against that, it assumes roughly convex and comparably sized clusters and understates the quality of partitions of elongated or irregular shape.

A.3.3 Variables Characterising the Clusters at $k = 3$

Table A.7 lists the value items on which the cluster profiles of Section 4.2 rest. For each cluster it reports the items selected by the one-vs-rest LASSO under at least two of the four clustering configurations, the configurations selecting them, and the geometric mean of the odds ratios from the unpenalised refit. These are the items that make the three-cluster description consistent across methods: an item shared by three or four configurations describes the cluster however the partition is drawn.

Table A.7: Variables selected by at least two clustering methods, by cluster ($k = 3$).

m	Var.	Question	Selected by	Mean OR
Cluster 1 — Disaffected				
4	Q165	Do you believe in God?	PC2+PC5+UMAP+PAM	1.870
3	Q57	Can most people be trusted, or do you need to be very careful?	PC2+UMAP+PAM	2.208
3	Q55	Gone without a safe shelter (last 12 months)	PC2+PC5+UMAP	0.373
3	Q81	Confidence in: charitable or humanitarian organisations	PC2+PC5+PAM	1.750
3	Q69	Confidence in: the police	PC2+UMAP+PAM	1.888
3	Q15	Child should learn: religious faith	PC2+UMAP+PAM	0.364
3	Q63	Trust: people of another nationality	PC2+PC5+UMAP	1.827
3	Q134	Police or military interfere with people's private life	PC2+PC5+UMAP	0.470
3	Q137	Street violence and fights in your neighbourhood	PC2+PC5+UMAP	0.537
2	Q25	Would not like as neighbour: unmarried couples living together	UMAP+PAM	0.287
2	Q18	Would not like as neighbour: drug addicts	PC2+PC5	0.270
2	Q122	Immigration fills important job vacancies	PC5+UMAP	0.555
2	Q60	Trust: people you know personally	PC5+UMAP	1.981
2	Q51	Gone without enough food to eat (last 12 months)	PC2+UMAP	0.503
2	Q88	Confidence in: the World Health Organization	PC5+UMAP	1.638
2	Q4	How important is politics in your life?	UMAP+PAM	1.474
2	Q76	Confidence in: elections	PC2+UMAP	1.838

continued on next page

Table A.7 – continued

m	Var.	Question	Selected by	Mean OR
2	Q119	On the whole, women are less corrupt than men	PC5+PAM	0.545
2	Q131	How secure do you feel these days?	PC2+PC5	2.024
2	Q66	Confidence in: the press	PC2+PC5	1.527
2	Q74	Confidence in: the civil service	PC5+PAM	1.752
Cluster 2 — Secular				
4	Q167	Do you believe in Hell?	PC2+PC5+UMAP+PAM	0.362
4	Q168	Do you believe in Heaven?	PC2+PC5+UMAP+PAM	0.438
3	Q126	Immigration increases the risks of terrorism	PC2+PC5+PAM	0.468
3	Q11	Child should learn: imagination	PC2+PC5+UMAP	2.238
3	Q130	What should the government do about people coming here to work?	PC2+PC5+UMAP	0.364
3	Q82	Confidence in: regional organisation (e.g. NAFTA)	PC2+PC5+PAM	0.660
3	Q169	Whenever science and religion conflict, religion is always right	PC2+PC5+PAM	3.651
3	Q111	Environmental protection vs. economic growth	PC2+UMAP+PAM	2.325
3	Q88	Confidence in: the World Health Organization	PC2+PC5+UMAP	0.462
3	Q29	On the whole, men make better political leaders than women do	PC2+PC5+PAM	2.366
2	Q154	Most important priority — *maintaining order*	PC2+UMAP	0.463
2	Q154	Most important priority — *protecting freedom of speech*	UMAP+PAM	1.683
2	Q15	Child should learn: religious faith	PC2+PC5	0.142
2	Q165	Do you believe in God?	UMAP+PAM	0.394
2	Q145	Has someone in your family been the victim of a crime?	PC2+PC5	0.527
2	Q55	Gone without a safe shelter (last 12 months)	PC5+UMAP	4.037
2	Q127	Immigration offers people from poor countries a better living	PC2+PC5	1.964
2	Q173	Religious person, not religious, or atheist?	PC2+UMAP	2.406
2	Q89	Confidence in: the World Trade Organization	UMAP+PAM	0.597
2	Q86	Confidence in: NATO	PC5+PAM	0.611
Cluster 3 — Religious				
4	Q126	Immigration increases the risks of terrorism	PC2+PC5+UMAP+PAM	1.944
4	Q167	Do you believe in Hell?	PC2+PC5+UMAP+PAM	2.364
4	Q71	Confidence in: the government	PC2+PC5+UMAP+PAM	0.544
4	Q45	Greater respect for authority: good, don't mind, or bad?	PC2+PC5+UMAP+PAM	0.574
4	Q154	Most important priority — *maintaining order*	PC2+PC5+UMAP+PAM	1.613
3	Q165	Do you believe in God?	PC2+PC5+UMAP	3.292
3	Q173	Religious person, not religious, or atheist?	PC2+PC5+UMAP	0.368
3	Q64	Confidence in: the churches	PC2+PC5+PAM	0.459
3	Q168	Do you believe in Heaven?	PC2+PC5+PAM	1.819
2	Q17	Child should learn: obedience	UMAP+PAM	1.845
2	Q69	Confidence in: the police	PC2+UMAP	0.629
2	Q116	Corruption: civil service providers	PC2+PC5	0.567
2	Q134	Police or military interfere with people's private life	PC5+UMAP	1.676
2	Q137	Street violence and fights in your neighbourhood	PC2+PC5	1.711
2	Q144	Have you been the victim of a crime during the past year?	PC2+PC5	0.426
2	Q77	Confidence in: major companies	PC2+PC5	0.683
2	Q30	A university education is more important for a boy than for a girl	PC5+UMAP	0.571
2	Q157	Most important societal goal (second choice)	PC2+PC5	1.643

Notes: m is the number of methods selecting the variable. Mean OR is the geometric mean of the odds ratios across those methods only, computed on the unpenalised refit; values below 1 indicate a negative association with cluster membership. Method labels are abbreviated (PC2, PC5, UMAP, PAM). Shaded rows are selected by all four methods. Ordinal variables are scored per step along the scale. Variables selected by a single method are omitted (28, 29 and 30 for clusters 1, 2 and 3 respectively). Signs are concordant across methods for every entry except Q154, reported separately by contrast.

A.4 Density-Based Clustering across Embedding Dimensions

A.4.1 Geometry of the Unassigned Respondents

Minimum Spanning Tree The minimum spanning tree on the mutual reachability distance — constructed as described in Section 3.3.2 — shows the retained clusters as dense knots at the tips of the branches, and the noise points as the branches joining them. Ninety-eight per cent of the splits in the tree generate noise points rather than clusters.[1]

A.4.2 Why so many noise points

The key is the distribution of the data: there is no actual cluster structure. As the distribution of pairwise Gower dissimilarities already showed (Figure 4.1), there is only one peak, meaning that the observed features describe a continuous differentiation rather than separated groups.

The single-linkage rule that builds the minimum spanning tree merges, at each step, the two components whose closest pair of points is nearest. Its weakness is *chaining*: if even a thin trail of points connects two dense regions, the rule walks along it step by step — each point almost touching the next — and fuses the two regions into one cluster. A single filament of points is enough to collapse two distinct modes.

The mutual reachability distance corrects this by reweighting exactly those filaments. It replaces the true distance with $d_{\text{mreach}}(a, b) = \max\{d_{\text{core}}(a), d_{\text{core}}(b), d(a, b)\}$, where the core distance is large not where points are few, but where they are not *concentrated* — where there is no local density peak. Points that form the connecting trails between modes live in precisely such unconcentrated regions, so their edges are lifted to at least their core distance and become heavy. The minimum spanning tree no longer uses them first, and the two modes stay separated until the higher levels of the hierarchy.

This is what the central region shows. It is not sparsely populated — it holds many respondents — but it has no density peak of its own: it is the plateau on which the peripheral modes rest. Its edges are therefore heavy in mutual reachability, the tree cuts them early (low λ_{death}), and the clusters that survive are the extremities of the continuum rather than distinct islands. The centre is dense in count but not separated in structure, which is why HDBSCAN assigns it to noise. Table A.8 reports, for each embedding dimensionality, the noise share and the share of respondents whose membership probability is below 0.5.

Table A.8: Primary HDBSCAN run ($\texttt{minPts} = 25$): clusters recovered, noise share, and share of respondents with membership probability below 0.5

UMAP dimension	Clusters	Noise (%)	Weak assignment (%)
2d	3	27.5	99.3
5d	6	68.7	98.3
20d	5	65.3	99.4
50d	5	66.4	99.6

A.4.3 Parameter Sensitivity

The `minPts` Grid Within the informative range of `minPts` the number of clusters varies from two to nine and changes with both the parameter and the embedding dimensionality. Two regions of the grid are degenerate. At $\texttt{minPts} = 5$ the algorithm returns exactly two clusters with almost no noise at every dimensionality: so small a neighbourhood makes essentially the whole cloud dense enough, and the near-zero noise reflects under-clustering rather than separation. At $\texttt{minPts} = 10$ it over-fragments, returning 39 clusters in two dimensions and 19 in fifty. The interpretable regime is $\texttt{minPts} \in [15, 30]$, where the noise share settles at 65–69% for $d \geq 5$.

[1] An interactive version of the graph is available at `https://rpubs.com/Regle/1447119`.

Table A.9: Adjusted Rand Index between primary partitions (noise included). $2d$ is an outlier; $5d/20d/50d$ are mutually consistent.

	$2d$	$5d$	$20d$	$50d$
$2d$	1.000	0.227	0.197	0.213
$5d$	0.227	1.000	0.710	0.712
$20d$	0.197	0.710	1.000	0.739
$50d$	0.213	0.712	0.739	1.000

Agreement across Embedding Dimensionalities The two-dimensional embedding disagrees with all the others ($\mathrm{ARI} \approx 0.20$), while the five-, twenty- and fifty-dimensional partitions agree at 0.71–0.74 (Table A.9). The same dense pockets are therefore recovered consistently at $d \geq 5$.

A.5 Value Groups as Coalitions

A.5.1 The Fractional Hedonic Game

Definitions Let $N = \{1, \ldots, n\}$ be the set of $n = 2{,}579$ respondents. A non-empty subset $H \subseteq N$ is a *coalition*; a partition $\pi = \{H_1, \ldots, H_K\}$ splits N into disjoint exhaustive coalitions, and $\pi(i)$ denotes the coalition containing respondent i. In a hedonic game each player ranks the coalitions she could join through a preference relation $\succeq_i$ depending only on the identity of her co-members. Respondent i *Nash-blocks* π if there exists $H \in \pi \cup \{\emptyset\}$ with $H \cup \{i\} \succ_i \pi(i)$: she strictly gains by leaving her coalition for another existing one, or by standing alone. A partition is *Nash-stable* when no player Nash-blocks it.

Utility Induced by the Gower Dissimilarity The game is a *fractional* hedonic game, in which the payoff from a coalition is the average value of its members. Writing $d_G(i,j) \in [0,1]$ for the Gower dissimilarity on the 198 value items, similarity is its complement,

$$v_i(j) = 1 - d_G(i,j) \in [0,1], \qquad v_i(i) = 0, \tag{A.3}$$

so that value-alike respondents are worth more to one another. The utility of i in a coalition $C \ni i$ is her mean similarity to its members,

$$u_i(C) = \frac{1}{|C|} \sum_{j \in C} v_i(j), \tag{A.4}$$

and standing alone yields $u_i(\{i\}) = 0$, the smallest attainable value. Substituting (A.3) into (A.4) and writing $s = |\pi(i)|$,

$$u_i\big(\pi(i)\big) = \frac{(s-1) - \sum_{j \in \pi(i)} d_G(i,j)}{s}, \qquad u_i\big(C \cup \{i\}\big) = \frac{|C| - \sum_{j \in C} d_G(i,j)}{|C|+1}, \tag{A.5}$$

the left expression being what i obtains by staying and the right what she would obtain by joining another coalition $C \neq \pi(i)$, which then has size $|C|+1$. The self-term drops out since $d_G(i,i) = 0$.

Deviators and the Status of the Criterion Respondent i has a *profitable deviation* in π if moving unilaterally strictly raises her utility:

$$\max\Big\{ \max_{C \in \pi,\, C \neq \pi(i)} u_i\big(C \cup \{i\}\big),\ 0 \Big\} \ > \ u_i\big(\pi(i)\big). \tag{A.6}$$

Let $D(\pi)$ be the number of respondents with a profitable deviation. A partition is Nash-stable exactly when $D(\pi) = 0$. $D(\pi)$ is reported as a diagnostic and not as a selection rule: ranking partitions by $D(\pi)$ would select the coarsest one mechanically, since fewer boundaries leave fewer border-dwelling respondents.

A.5.2 Results

Deviators across Configurations and k No candidate partition is Nash-stable: $D(\pi) > 0$ everywhere. The least-contested configuration is PAM at $k = 2$, which still leaves 277 respondents, one in ten, with a profitable deviation. The share rises with k in every configuration, though not monotonically: for KM-PC5 it goes from 14.4% at $k = 2$ to 26.9% at $k = 6$, with a dip at $k = 4$ (Table A.10).

Table A.10: Deviators and aggregate deviation gains under the fractional hedonic game, by configuration and number of clusters ($N = 2{,}579$). No configuration at any k is Nash-stable, so the stability column is omitted. The HDBSCAN row is computed only over the $1{,}869$ respondents it assigns, the remaining 710 being noise, and its share is therefore not comparable with the others.

Configuration	k	Assigned	Deviators	Share (%)	Aggregate gain
PAM (Gower)	2	2,579	277	10.7	3.18
KM-UMAP	2	2,579	320	12.4	3.28
KM-PC2	2	2,579	351	13.6	3.78
KM-PC5	2	2,579	371	14.4	4.09
HDBSCAN (UMAP)	3	1,869	284	15.2	6.18
KM-UMAP	3	2,579	579	22.5	10.55
KM-PC2	3	2,579	599	23.2	10.88
PAM (Gower)	3	2,579	606	23.5	7.61
KM-PC5	3	2,579	621	24.1	13.52
KM-PC5	4	2,579	579	22.5	10.66
KM-UMAP	4	2,579	676	26.2	12.39
PAM (Gower)	4	2,579	682	26.4	9.28
KM-PC2	4	2,579	686	26.6	12.09
KM-PC5	5	2,579	668	25.9	12.44
KM-UMAP	5	2,579	712	27.6	13.25
PAM (Gower)	5	2,579	730	28.3	8.97
KM-PC2	5	2,579	854	33.1	13.77
KM-PC5	6	2,579	694	26.9	11.94
PAM (Gower)	6	2,579	710	27.5	9.02
KM-UMAP	6	2,579	896	34.7	15.89
KM-PC2	6	2,579	960	37.2	14.30

A.6 Recovery of Cluster Membership at Every k

Section 4.3 reports the recovery exercise at $k = 3$. The tables below repeat it for every number of clusters from two to six. Tables A.11, A.12 and A.13 give accuracy and κ_{emp} for the political block, the sociodemographic block and their union; Table A.14 compares the three blocks directly. The rows at $k = 3$ coincide with Table 4.4.

Table A.11: Out-of-sample recovery of cluster membership from the **political block** (60 variables, 256 dummies; 5-fold cross-validation). Accuracy is reported against the uniform baseline $1/k$; κ_{emp} corrects against the observed cluster marginals and is the measure comparable across k.

			Accuracy		κ_{emp}	
k	Configuration	Baseline $1/k$	LASSO	RF	LASSO	RF
2	KM-PC2	50.0%	84.3%	84.4%	0.663	0.659
	KM-PC5	50.0%	85.1%	85.3%	0.681	0.677
	KM-UMAP	50.0%	81.6%	82.4%	0.619	0.630
	PAM	50.0%	77.9%	79.3%	0.514	0.542
3	KM-PC2	33.3%	69.2%	68.7%	0.532	0.524
	KM-PC5	33.3%	71.0%	71.3%	0.556	0.560
	KM-UMAP	33.3%	69.9%	70.4%	0.538	0.545
	PAM	33.3%	60.3%	62.3%	0.404	0.433
4	KM-PC2	25.0%	59.2%	60.0%	0.456	0.466
	KM-PC5	25.0%	66.8%	64.6%	0.514	0.474
	KM-UMAP	25.0%	59.3%	59.0%	0.456	0.452
	PAM	25.0%	50.8%	52.5%	0.336	0.360
5	KM-PC2	20.0%	53.3%	53.0%	0.410	0.403
	KM-PC5	20.0%	58.5%	57.8%	0.461	0.450
	KM-UMAP	20.0%	54.7%	54.9%	0.420	0.420
	PAM	20.0%	46.7%	47.3%	0.314	0.320
6	KM-PC2	16.7%	47.3%	48.9%	0.358	0.374
	KM-PC5	16.7%	53.1%	52.7%	0.420	0.413
	KM-UMAP	16.7%	50.2%	48.9%	0.397	0.381
	PAM	16.7%	42.4%	43.4%	0.284	0.293

Table A.12: Out-of-sample recovery of cluster membership from the **sociodemographic block** (27 variables, 196 dummies; 5-fold cross-validation). Recovery is well above chance but consistently weaker than from the political block (Table A.11): at $k = 2$, κ_{emp} reaches 0.39 against 0.68.

			Accuracy		κ_{emp}	
k	Configuration	Baseline $1/k$	LASSO	RF	LASSO	RF
2	KM-PC2	50.0%	71.8%	72.3%	0.373	0.385
	KM-PC5	50.0%	72.5%	72.2%	0.387	0.380
	KM-UMAP	50.0%	70.6%	70.2%	0.383	0.373
	PAM	50.0%	70.9%	70.5%	0.332	0.334
3	KM-PC2	33.3%	60.6%	60.1%	0.401	0.393
	KM-PC5	33.3%	62.0%	62.2%	0.416	0.419
	KM-UMAP	33.3%	60.6%	60.4%	0.396	0.392
	PAM	33.3%	53.8%	52.6%	0.306	0.288
4	KM-PC2	25.0%	48.4%	47.7%	0.310	0.298
	KM-PC5	25.0%	57.2%	57.5%	0.368	0.373
	KM-UMAP	25.0%	50.6%	50.5%	0.341	0.339
	PAM	25.0%	45.9%	44.5%	0.271	0.253
5	KM-PC2	20.0%	42.5%	41.8%	0.271	0.263
	KM-PC5	20.0%	47.4%	46.1%	0.313	0.297
	KM-UMAP	20.0%	45.8%	44.8%	0.307	0.294
	PAM	20.0%	40.3%	40.6%	0.224	0.229
6	KM-PC2	16.7%	38.8%	36.0%	0.243	0.212
	KM-PC5	16.7%	44.3%	42.1%	0.311	0.283
	KM-UMAP	16.7%	42.1%	41.2%	0.297	0.288
	PAM	16.7%	36.5%	36.1%	0.204	0.200

Table A.13: Out-of-sample recovery of cluster membership from the **combined political and sociodemographic block** (87 variables, 452 dummies; 5-fold cross-validation). Accuracy is reported against the uniform baseline $1/k$; κ_{emp} corrects against the observed cluster marginals.

			Accuracy		κ_{emp}	
k	Configuration	Baseline $1/k$	LASSO	RF	LASSO	RF
2	KM-PC2	50.0%	85.7%	85.3%	0.692	0.677
	KM-PC5	50.0%	85.5%	86.1%	0.686	0.696
	KM-UMAP	50.0%	81.8%	82.9%	0.624	0.642
	PAM	50.0%	79.4%	79.4%	0.549	0.544
3	KM-PC2	33.3%	71.6%	71.2%	0.569	0.561
	KM-PC5	33.3%	73.2%	73.5%	0.591	0.594
	KM-UMAP	33.3%	72.6%	71.8%	0.580	0.567
	PAM	33.3%	62.5%	62.6%	0.437	0.437
4	KM-PC2	25.0%	60.9%	62.7%	0.477	0.502
	KM-PC5	25.0%	68.6%	66.8%	0.541	0.507
	KM-UMAP	25.0%	61.8%	62.4%	0.490	0.498
	PAM	25.0%	52.7%	54.4%	0.363	0.385
5	KM-PC2	20.0%	55.5%	55.8%	0.438	0.439
	KM-PC5	20.0%	60.4%	59.6%	0.484	0.473
	KM-UMAP	20.0%	58.6%	57.0%	0.473	0.448
	PAM	20.0%	48.9%	48.7%	0.342	0.337
6	KM-PC2	16.7%	52.0%	51.3%	0.414	0.402
	KM-PC5	16.7%	57.9%	56.4%	0.480	0.459
	KM-UMAP	16.7%	53.4%	52.1%	0.436	0.420
	PAM	16.7%	44.4%	44.6%	0.309	0.308

Table A.14: Recovery by predictor block, all values of k. Entries are κ_{emp} under LASSO-penalised multinomial regression. The last column is the gain of the combined block over the better of the two taken alone, which is positive everywhere but small: political attitudes and social location describe largely the same variation.

k	Configuration	Political	Sociodem.	Both	Gain over best single
2	KM-PC2	0.663	0.373	**0.692**	+0.029
	KM-PC5	0.681	0.387	**0.686**	+0.005
	KM-UMAP	0.619	0.383	**0.624**	+0.005
	PAM	0.514	0.332	**0.549**	+0.035
3	KM-PC2	0.532	0.401	**0.569**	+0.037
	KM-PC5	0.556	0.416	**0.591**	+0.035
	KM-UMAP	0.538	0.396	**0.580**	+0.042
	PAM	0.404	0.306	**0.437**	+0.033
4	KM-PC2	0.456	0.310	**0.477**	+0.021
	KM-PC5	0.514	0.368	**0.541**	+0.027
	KM-UMAP	0.456	0.341	**0.490**	+0.034
	PAM	0.336	0.271	**0.363**	+0.027
5	KM-PC2	0.410	0.271	**0.438**	+0.028
	KM-PC5	0.461	0.313	**0.484**	+0.023
	KM-UMAP	0.420	0.307	**0.473**	+0.053
	PAM	0.314	0.224	**0.342**	+0.028
6	KM-PC2	0.358	0.243	**0.414**	+0.056
	KM-PC5	0.420	0.311	**0.480**	+0.060
	KM-UMAP	0.397	0.297	**0.436**	+0.039
	PAM	0.284	0.204	**0.309**	+0.025